Bayesian Cybersecurity



THE LFP BATTERY ANALYTICS APP · A BUSINESS GUIDE

Xtreme Analytics

Battery analytics and battery cybersecurity
for xtreme battery business problems

FOR LFP BATTERIES

Health, remaining life, warranty, resale, heat, balance — and Stuxnet-style attack detection. Start with your question. The book points you to the screen that answers it.

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Open the live app → **batteryanalytics.bayesiancybersecurity.com**

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Battery analytics and battery cybersecurity for LFP packs

THE LFP BATTERY ANALYTICS APP

Rahul Modak · Dr Rahul Walawalkar

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Please read this first

IMPORTANT DISCLAIMER — EXPERIMENTAL TECHNOLOGY

The numbers in this book and in the app come from the Modak–Walawalkar (MW) physics AI, which is experimental technology. AI can make mistakes. Results are estimates, not measurements, and may be wrong. The service is provided without warranty of any kind, express or implied. Bayesian Cybersecurity and its authors accept no liability of any kind, to the fullest extent permitted by law, for any loss, damage or decision arising from its use. Use it entirely at your own risk, and consult a qualified battery chemist or engineer before relying on any result.

Validation so far. On two public lab datasets (Sandia, 18 cells; HUST, 77 cells; A123 LFP 18650 cells), scored on cells it had never seen, the engine forecast battery health within 2.4–3.4 percentage points (typical error), against 18–21 points for the plain physics formula alone. That is encouraging, but it is lab data on one cell type. Field packs and other chemistries have not been validated.

In plain words. It is a smart second opinion, built and tested for LFP batteries. Use it to decide where to look and what to plan. It does not replace a lab test, and it is not a safety system: it cannot detect thermal runaway, internal shorts, venting or fire, and it must never override a BMS warning.

About the screenshots

Two packs appear in this book. Both are 16-cell LFP packs.

- **The Quick Analysis pack** (Chapter 2) is the one-page summary view on a 17.6-hour log.
- **The demo pack** (Chapters 3–13) is synthetic. Its CAN log came from our own data generator. We know exactly what we planted in it — which cell is weak, which sensor is dead — so we can show what the app should find. A real pack will look messier. That is why the app shows ranges, not single numbers.

About this edition

Third edition. Adds Part 4, on Stuxnet-style attacks on batteries and how physics-level checks detect what network monitors miss.

The first edition walked through the app screen by screen. Readers told us the obvious: nobody buys a screen. They buy an answer to a business question. So this edition is organised around the questions. Each chapter opens with a question, says who asks it, gives the quick answer, and only then shows the screens that go deeper.

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START HERE

How to use this book: find your question

Thirteen questions people actually ask about batteries, and where each one is answered.

Built for LFP. Xtreme Analytics is an LFP battery analytics app. Every model, check and number in this book is built and tested for lithium iron phosphate (LFP) packs. Other chemistries are not supported yet.

Find your question in the left column. The Quick column says whether the one-page summary answers it. If you need more, go to the chapter.

The business question	Who usually asks	Quick page answers it?	Chapter
Is my battery healthy? The BMS says 100%. Is that true?	Owner, fleet manager, buyer	**Yes** · health tile	3
How long will it last? How many years and cycles are left?	Asset manager, CFO, lender	**Yes** · remaining-life tile	4
What warranty can I offer? What reserve should I hold?	OEM, pack maker, financier, insurer	**Partly** · usage + life tiles	5
Which cell is the problem? Do I send a technician?	Maintenance lead, service partner	**Yes** · cell grid + What to do	6
Is it a weak cell or just out of balance? How much energy am I losing?	Operations, technician	**Yes** · energy tile + balance note	7
Is it running hot? Where?	Maintenance, safety officer	**Partly** · temperature chart	8
Which operating habit should I change, and does it pay back?	Operations, fleet manager	**No** · Detailed only	9
What is this used pack worth? Can it go to second life?	Buyer, refurbisher, insurer, lender	**Partly** · health + usage tiles	10
Is the data honest? Sensor fault, wrong setting or tampering?	Security, auditor, insurer	**Partly** · checks flagged	11

Can I trust this number enough to put it in a contract?	Anyone quoting a number	**Partly** · ranges on every tile	12
How do I do this for 100 packs, in any log format?	Fleet owner, OEM service head	—	13
Could someone attack my battery Stuxnet-style? Would I know?	BESS owner, utility, insurer, security head	**Partly** · checks flagged	14, 15
I have the findings. What do I do, and by when?	Everyone	**Yes** · What to do	16

Who should read what

If you are…	Read
Owner or CEO with ten minutes	Chapter 2 (Quick Analysis), then Chapter 16 (playbook)
Finance, warranty, insurance, lending	Chapters 2, 4, 5, 10, 12
Fleet or asset manager	Chapters 2, 4, 6, 9, 13
Maintenance lead or technician	Chapters 6, 7, 8, 11, then 14
Battery engineer	Chapters 3, 6, 10, 12, 17
Security or audit	Chapters 11, 12, 14 and 15
Sceptical (good)	Chapters 1, 12 and 17

PART 1

The landscape

What today's battery analytics platforms do well, which questions they leave open, and why the MW approach can answer them.

CHAPTER 1

The battery analytics market, and the questions it leaves open

A fair look at the competition, and where Xtreme Analytics fits.

Battery analytics is now an established software category. A handful of well-funded firms, mostly in Europe and the US, serve large energy storage owners, EV makers and big fleets. They are good at what they do. This chapter says what that is, and which questions remain unanswered.

What the leading platforms do well

- **Continuous cloud monitoring.** They connect to a BESS site or a vehicle fleet and watch it around the clock.
- **Health, safety and performance dashboards.** State of health, availability, round-trip efficiency, early safety alerts and anomaly detection.
- **Warranty tracking.** Some track warranty conditions across a portfolio and warn of breaches.
- **Field experience.** Years of data across many sites, chemistries and vendors.
- **Accuracy guarantees.** At least one leader backs its health numbers with an insurance-backed guarantee.
- **Integrations.** APIs and data bridges into energy management and fleet systems.

How they are bought

They are sold as enterprise platforms. Prices are on request. Onboarding means connecting your data pipeline to theirs and tuning to your assets. Even vendors who call their onboarding fast speak of about four weeks, and of several months for others. Their health and degradation insights improve with months of history.

That model suits a 500 MWh BESS portfolio with a data team. It does not suit a battery-swapping operator with 300 e-rickshaw packs, a pack maker answering one warranty claim, a refurbisher buying a lot of used modules, or an insurer checking one incident log.

Built for the operators the market skips

Much of the world's LFP runs outside big BESS portfolios: e-rickshaws, two- and three-wheelers, battery swapping, telecom backup and small solar storage. In India and across the Global South, that means thousands of small operators. They run hundreds of small packs on thin margins, often with low-cost cells, mixed BMS formats and no data team.

As far as we know, we may be the only battery analytics platform built for these operators at scale. It needs no GPU, no integration project and no months of history — one CAN log in a browser. Our aim is to become the global LFP analytics platform, starting where LFP is worked hardest.

Questions the market leaves open

The question	Typical answer today	Xtreme Analytics
"I have one log file. What is this pack's condition today?"	Connect the asset; insights build up over weeks to months of history.	One 12–24 hour BMS / CAN log, in almost any column layout. Answer in minutes.
"Will it last N cycles? Give me a probability, not a number."	A state-of-health number and a remaining-life estimate.	A range on every forecast, plus the chance it lasts N cycles. Directly usable in warranty and loan terms (Chapters 4, 5).
"Is the data itself honest?"	Telemetry is mostly taken at face value. Anomaly detection looks for odd battery behaviour, not forged data.	Ten physics checks on the raw log. CAN has no passwords, but faked data still has to obey physics (Chapter 11).
"Is that difference real, or noise?"	Thresholds: red, amber, green.	A measurement floor and overlapping ranges. The app says "not distinguishable" or "NOT resolved" instead of guessing (Chapter 12).
"Which habit change pays back, and is the effect bigger than the uncertainty?"	Scenario tools, where offered, give a point answer.	The Virtual Lab calls a habit "resolved" only if its effect beats the forecast range (Chapter 9).
"Can I check the evidence myself?"	Accuracy claims based on the vendor's own data.	Validation on public lab datasets anyone can obtain. The full report is downloadable (Chapter 17).
"I'm small. Can I start today?"	Enterprise sale, integration project.	Upload in a browser. Up to 100 logs at once (Chapter 13).

Where the leaders are ahead

We say this plainly because you will find out anyway.

- **Field validation.** They have years of real packs. Our validation is on public lab data for one LFP cell type. Field packs are next.
- **Chemistries.** They cover several. We are built and tested for LFP.
- **Real-time safety.** They run continuous safety monitoring. We are not a safety system and never will be from a log file.
- **Guarantees and integrations.** Insurance-backed accuracy and deep EMS integrations take years to build.

> **Our honest position.** If you run a large BESS portfolio with a data team, a leading platform may be the right primary tool. Xtreme Analytics is then a cheap independent second opinion and a data-integrity check. If you have logs but no platform, it is the fastest way to a defensible answer.

Why the MW approach can answer these questions

Behind the app is the Modak–Walawalkar (MW) Framework, a probabilistic approach to battery physics. It replaces slow physics equations with a learned map of what a healthy LFP cell is allowed to do.

What MW does	So you get

Builds physics rules in: voltage stays in the chemistry's window, resistance falls as a cell warms, capacity never grows back.	Useful answers from one day of data. The rules do much of the work that months of history do elsewhere.
Is probabilistic from the ground up.	A range and a probability on every forecast, not just a single number.
Measures how far a cell sits from what it knows ("MW distance").	It tells you when it does not know. Few tools do.
Checks the log against ten physics laws before any forecast.	Data integrity and a cyber check come free with the health check.
Forces health to only fall (batteries do not heal).	No rosy surprises in a forecast.
Runs fast on ordinary computers.	Quick results in seconds; a fleet of logs in one batch.

Xtreme Analytics is a product built on the MW Framework, for LFP batteries. This book leaves the maths out.

PART 2

The two-minute answer

One page that answers the questions most people ask first: how healthy, how long, how much energy, how hard used, and what to do.

CHAPTER 2

Quick Analysis: the page that answers most questions

Upload one log. Read one page. Decide.

Upload a 12–24 hour BMS / CAN log and choose Quick. In seconds you get one page written for the person who pays for the battery, not the person who builds models. It answers five questions and ends with a short to-do list.

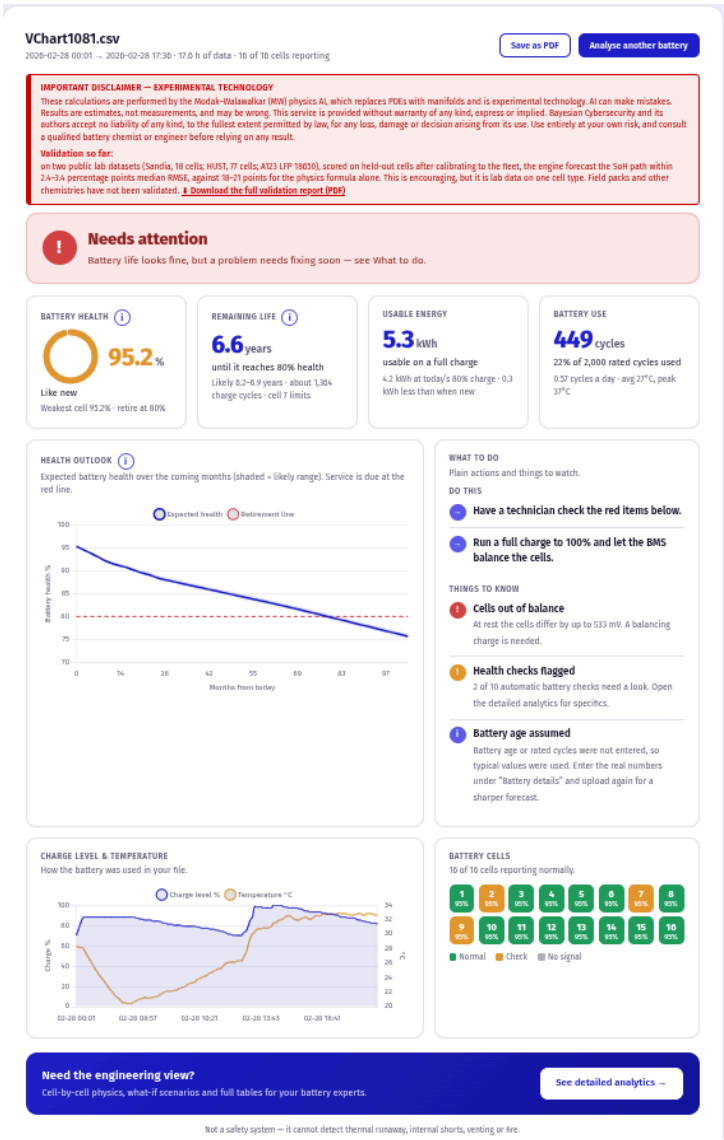


Figure 2.1 — The Quick Analysis page for a 16-cell LFP pack, 17.6 hours of data. The rest of this chapter reads it top to bottom.

Question 1 · Is it fine? The verdict



You see	One plain verdict. Here: **Needs attention** — battery life looks fine, but a problem needs fixing soon.
Read it as	The verdict separates two things owners often mix up: is the battery *ageing* badly, and is something *wrong* right now. Here ageing is fine; a fixable problem exists.
Decide	Go straight to "What to do" below. No replacement decision yet.

Questions 2–5 · Health, life, energy, use



Tile	This pack	The business question it answers
Battery health	**95.2%** "Like new". This is the weakest cell; the pack retires at 80%.	What is it worth today? A pack is only as good as its weakest cell, so the tile reports that cell.
Remaining life	**6.6 years** to 80% health. Likely 6.2–6.9 years, about 1,364 charge cycles. Cell 7 sets the limit.	When do I budget a replacement? How long can a loan or lease run?
Usable energy	**5.3 kWh** on a full charge. 4.2 kWh at today's 80% charge. 0.3 kWh less than when new.	What can it deliver now? What has ageing cost me in energy?
Battery use	**449 cycles**, 22% of 2,000 rated. 0.57 cycles a day; average 27 °C, peak 37 °C.	How hard has it been worked against its rating? Is the duty gentle or abusive?

Warranty in one line, from the tiles

The tiles together answer a question no single tile does. At this pack's duty:

- Cycles used so far: 449. Cycles to 80% health: about 1,364 more. So 80% is reached at about **1,813 cycles**.
- The rated life is 2,000 cycles. At 0.57 cycles a day, 2,000 cycles would take about **7.5 more years**.
- The forecast says 80% arrives in about **6.6 years** — before the rated cycles are used up.

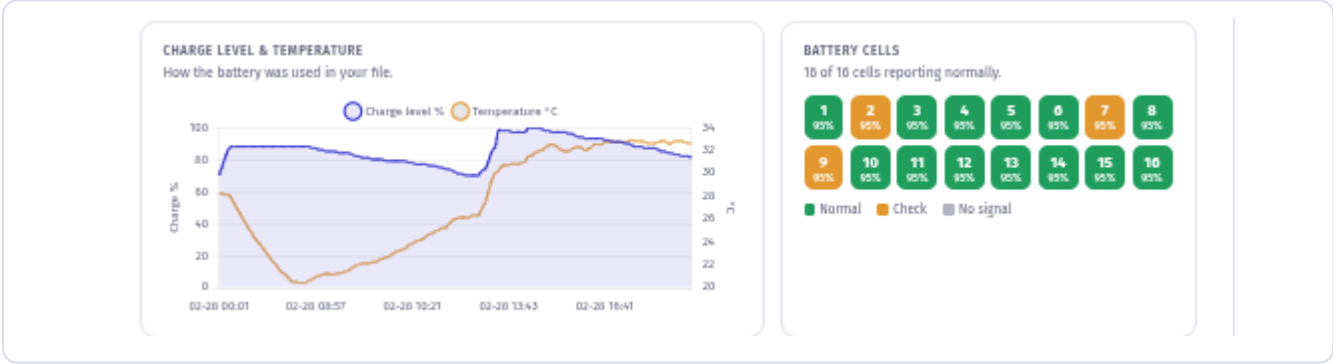
> **So what.** A warranty of "80% health for 2,000 cycles" on this pack, at this duty, carries real claim risk in years 6–7. A warranty of "80% for 6 years" has headroom. Chapter 5 turns this into a probability you can price. **Caveat:** this pack's age and rated cycles were assumed (see "Battery age assumed" below). Enter the real nameplate values and re-run before quoting any of this.

Question 6 · What is coming, and what do I do?



You see	Left: **Health outlook** — expected health over the coming months, a shaded likely range, and a red line at 80%. Right: **What to do**, in plain words.
Read it as	Service is due where the blue line meets the red line, here roughly six to seven years out. "Do this" lists actions. "Things to know" lists findings: cells out of balance by up to 533 mV at rest; 2 of 10 automatic health checks need a look; battery age was assumed.
Decide	Give "Do this" to the technician as written: check the flagged items, then a full charge to 100% so the BMS can balance the cells. Fill in the real age and rated cycles, and re-run.

Question 7 · How was it used, and which cells need a look?



You see	Left: charge level and temperature through the log. Right: all 16 cells. Green is normal; orange is "check"; grey would be "no signal".
Read it as	All 16 cells report. Cells 2, 7 and 9 are marked "check". The temperature rose by about 10 °C through the day while the pack sat near full charge.
Decide	Orange cells go on the technician's list. For why a cell is orange, tap See detailed analytics → Pack overview (Chapter 6).

Need more? Tap "See detailed analytics"



At the bottom of the Quick page is **See detailed analytics →**. It opens the same log in Detailed mode. No re-upload. Detailed mode has ten tabs across the top:



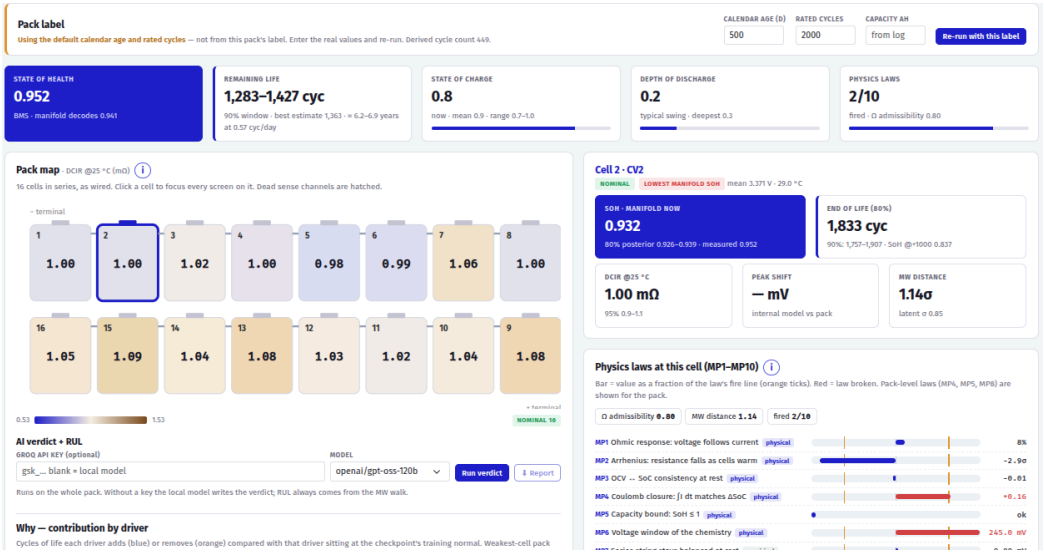
You never need all ten. Start from your business question and go to the tab that answers it.

Business question	Quick page	Go to this Detailed tab	Chapter
How healthy is it, really?	Health tile	Pack overview	3
How long will it last?	Life tile, outlook	Life forecast	4
What warranty can I offer?	Life + use tiles	Life forecast · Physics Lab	5
Which cell is the problem?	Cell grid	Pack overview · Cell internals	6
Weak cell or balancing job?	Balance note	Thermal & balance · Telemetry replay	7
Is it running hot?	Temperature line	Thermal & balance	8
Which habit should I change?	—	Capacity & usage · Virtual Lab · Physics Lab	9
What is this used pack worth?	Health + use tiles	Cell internals · Pack overview	10
Is the data honest?	"Checks flagged"	Pack overview · MW manifold & physics	11
Can I trust this number?	Ranges on tiles	MW manifold & physics · Data & conversion	12
How do I do this for a fleet?	—	Data & conversion · Fleet	13

The ten Detailed tabs, by business question

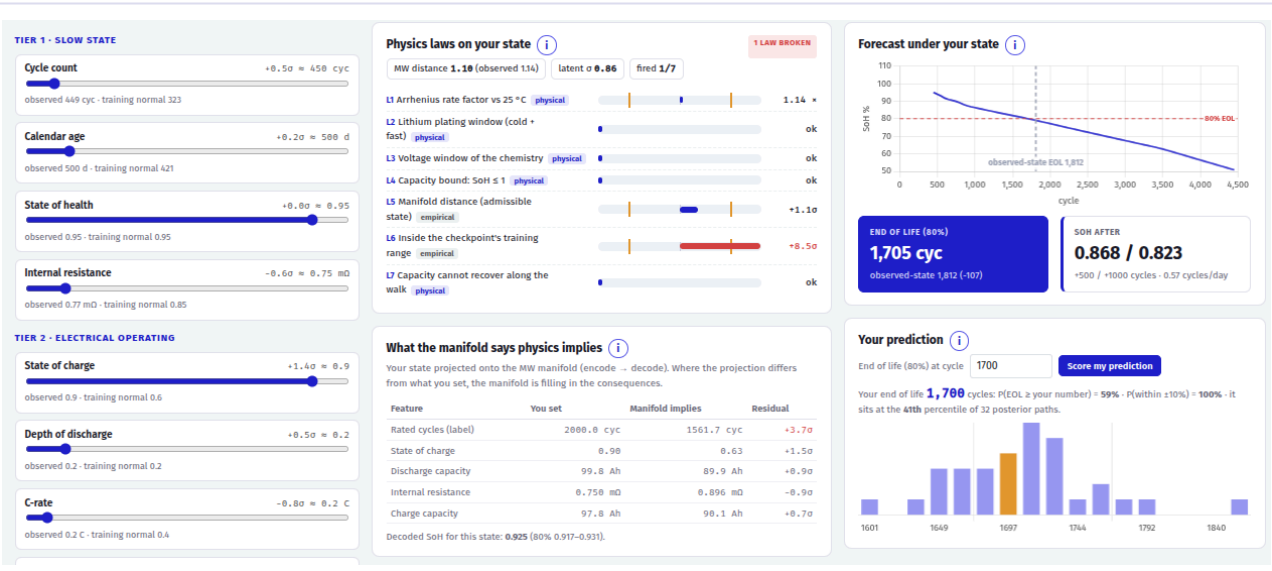
These screenshots are the same pack as the Quick page, opened with the button. Later chapters use a synthetic demo pack to show planted faults.

Tab 1 · Pack overview



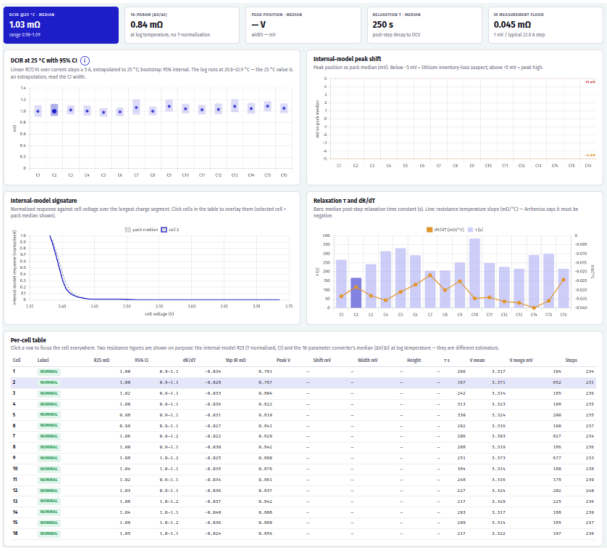
Answers	**Is it healthy? Which cell is the problem? Does the data hold together?**
This pack	BMS health 0.952, engine estimate 0.941. Remaining life 1,283–1,427 cycles (about 6.2–6.9 years at 0.57 cycles a day). 2 of 10 physics checks fired. Pack map with eight colour layers, cell card, ten checks, and what drives the life number.
Read more	Chapter 3, 6, 11

Tab 2 · Physics Lab



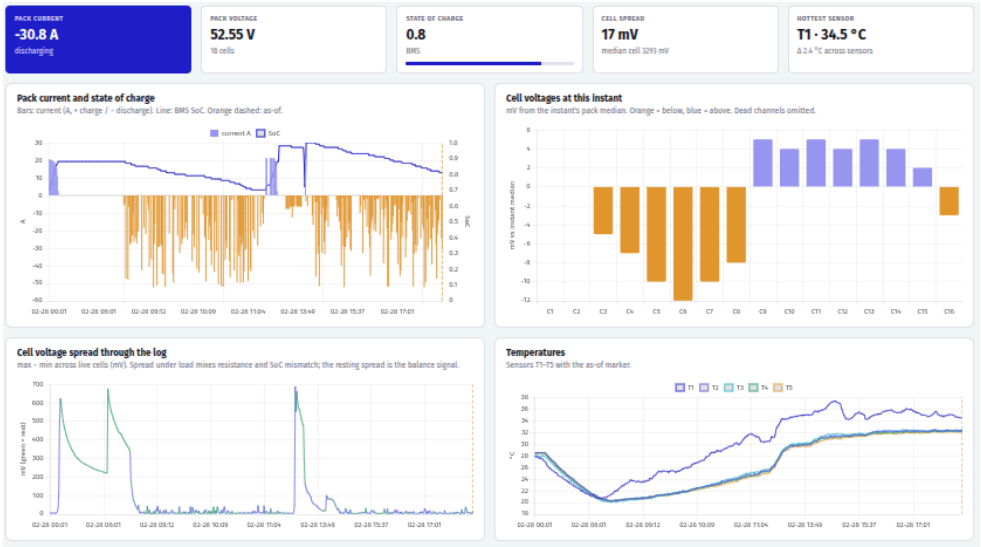
Answers	**What if we used it differently? Will it last N cycles?**
This pack	Set the duty with sliders. "Your prediction" at 1,700 cycles: 59% chance the pack lasts at least that long. Forecast under the set state: 1,705 cycles.
Read more	Chapter 5, 9

Tab 3 · Cell internals



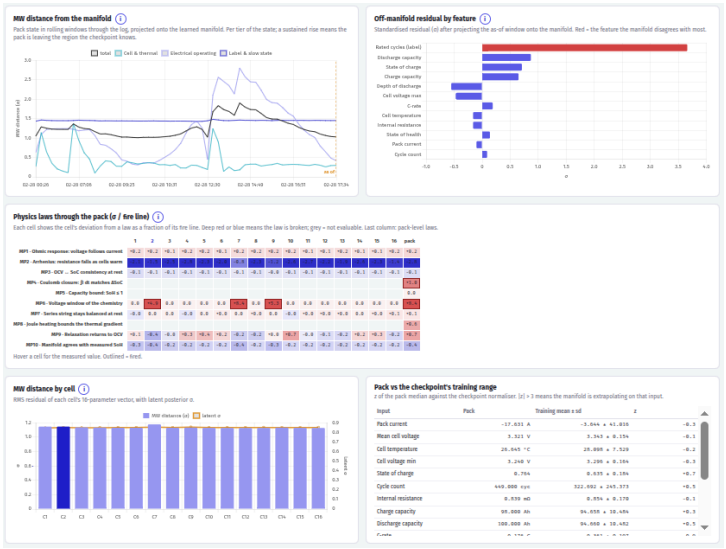
Answers	**What kind of wear is it? Which cells do I grade down?**
This pack	Resistance at 25 °C: median 1.03 mΩ, range 0.98–1.09. All 16 cells NOMINAL. Settling time 250 s. Per-cell table as the audit trail.
Read more	Chapter 6, 10

Tab 4 · Telemetry replay



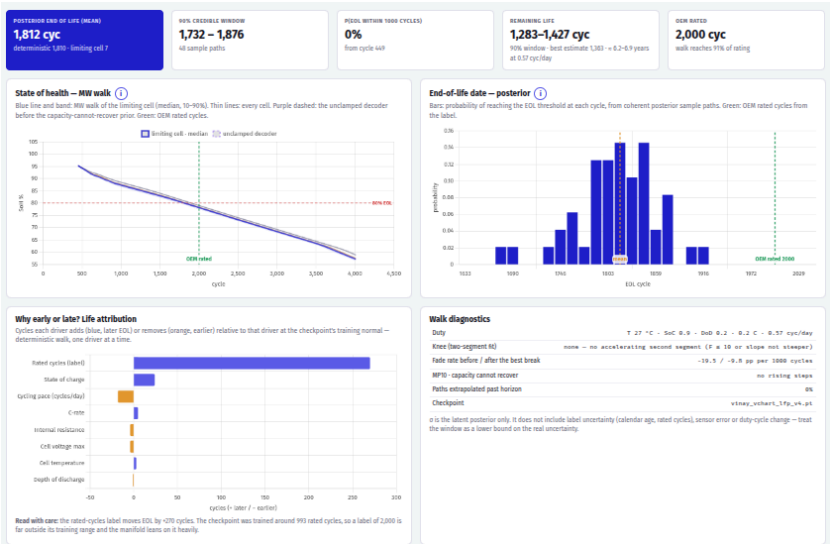
Answers	**What actually happened, and when? Can I show the evidence?**
This pack	Slide to any moment: current, voltage, charge level, cell spread, hottest sensor. Cell spread through the log shows where imbalance appears.
Read more	Chapter 7, 11

Tab 5 · MW manifold & physics



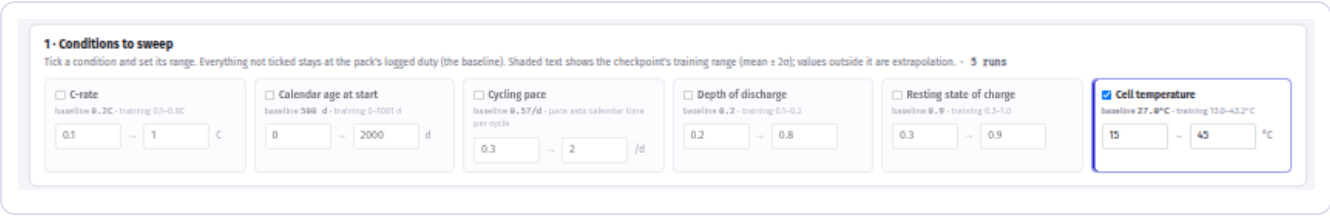
Answers	**Can I trust the number? Is the data consistent with physics?**
This pack	MW distance stays below 2: familiar ground. The biggest misfit is the rated-cycles label, not the battery — a reason to enter the real label. Ten checks across every cell.
Read more	Chapter 11, 12

Tab 6 · Life forecast



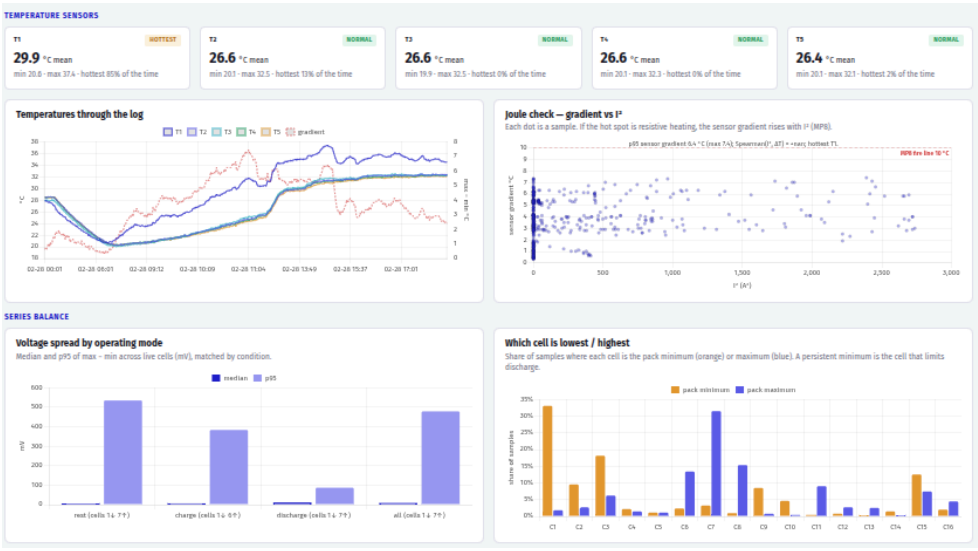
Answers	**How long will it last? What warranty can I offer?**
This pack	End of life (80%) at 1,812 cycles on average; 90% window 1,732–1,876. 0% chance of reaching end of life within 1,000 cycles. Reaches 91% of the 2,000-cycle rating. No knee found.
Read more	Chapter 4, 5

Tab 7 · Virtual Lab



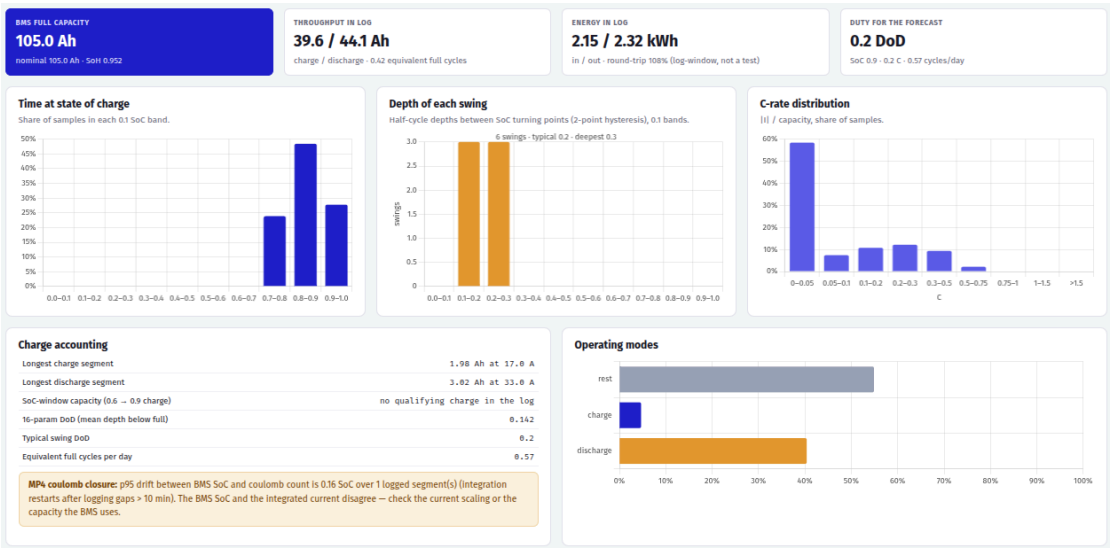
Answers	**Which operating habit should I change, and does it pay?**
This pack	Tick conditions to sweep: charge rate, starting age, cycling pace, depth of discharge, resting charge level, temperature. Each lever is ranked in cycles and years, and marked "resolved" only if it beats the uncertainty. The Virtual Lab has reproduced the published findings of leading peer-reviewed battery researchers on their own experimental data.
Read more	Chapter 9

Tab 8 · Thermal & balance



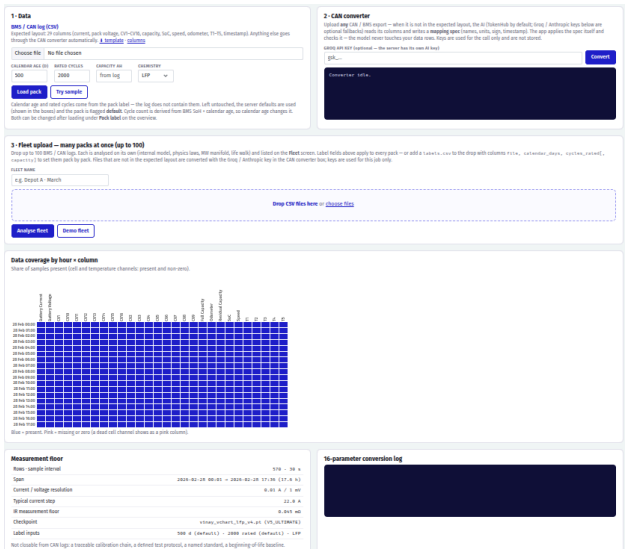
Answers	**Is it running hot, and where? Weak cell or balancing job?**
This pack	T1 is the hottest sensor 85% of the time (mean 29.9 °C vs about 26.5 °C for the rest). Resting spread is large: a balancing job first.
Read more	Chapter 7, 8

Tab 9 · Capacity & usage



Answers	**How hard is the pack worked? What duty does the forecast assume?**
This pack	0.57 equivalent full cycles a day, 0.2 depth, mostly above 70% charge. Check MP4 flags that the BMS charge level and the measured current disagree — check the current scaling.
Read more	Chapter 5, 9, 11

Tab 10 · Data & conversion



Answers	**Can it read my logs? Can I run a fleet? What can the sensors see?**
This pack	Upload any CAN / BMS export; the converter maps the columns. Fleet upload of up to 100 logs. Coverage heatmap and measurement floor (0.045 mΩ).
Read more	Chapter 12, 13

> **Rule of thumb.** Quick for every pack. See detailed analytics for the packs where money is at stake. Save as PDF builds a shareable report from either.

PART 3

The business problems

Eleven chapters, one question each. Every chapter gives the quick answer first, then the screens that go deeper, then what the tool cannot tell you.

CHAPTER 3

Is my battery healthier, or sicker, than the BMS says?

Why an LFP BMS can read 100% while a cell quietly fades.

Who asks: owners, fleet managers, used-pack buyers, lenders.

QUICK ANSWER

The health tile (Chapter 2) gives the weakest cell's health. Detailed mode puts the BMS number and the engine's own estimate side by side. A gap between them is a finding in itself.

Where in the app: Quick page → *See detailed analytics* → **Pack overview** (pack label, state of health, Manifold SoH layer, cell card, check MP10).

Why the BMS can be wrong on LFP

Most lithium-ion cells have a voltage that slopes as they charge. LFP does not. Its voltage sits on a flat plateau for most of the charge. A BMS that reasons from voltage sees almost nothing change as the cell ages. Two cells with very different remaining capacity can read the same.

The clues are elsewhere: how big the voltage jump is when current changes, how fast voltage settles, where the cell takes in charge fastest, how it spreads heat, and how it drifts from its neighbours. All of it is in a day of ordinary CAN data.

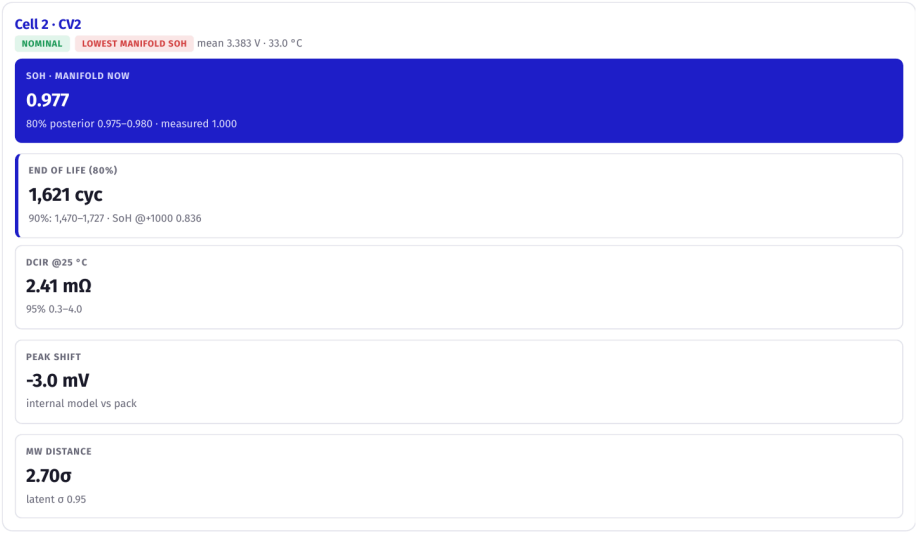
What wears a cell out	In plain words	Where it shows in the data
SEI growth	A film on the anode thickens. Uses up lithium, adds resistance.	Slow capacity loss, slowly rising resistance. Faster when hot or stored full.
Lithium loss	Lithium gets locked away for good. The main fade type in LFP.	The charging peak moves. Capacity drops with little change in resistance.
Active material loss	Part of the electrode stops working.	The charging peak gets shorter.
Resistance growth	The cell fights current harder.	Bigger voltage jumps, slower settling, more heat.
Lithium plating	Metal lithium deposits, mostly when fast-charged in the cold.	Fade speeds up (the "knee").
Heat imbalance	Uneven cooling, loose joints.	One sensor always hottest.
Internal short	A leak path inside a cell.	A cell that sags at rest. A safety matter, outside what this tool can certify.

Two health numbers, on purpose



You see	The pack label (age, rated cycles, capacity) and state of health: 1.000 from the BMS, 0.978 from the engine.
Read it as	Age and rated cycles are not in a CAN log. Until you enter them, the app uses defaults and marks the pack "default".
Decide	Enter the nameplate values before quoting anything. A BMS–engine gap is a reason to audit the BMS health counter.

One cell, with its range



You see	Cell 2: health 0.977 (likely 0.975–0.980; BMS says 1.000); end of life at 1,621 cycles (likely 1,470–1,727); resistance 2.41 mΩ; peak shift −3.0 mV; MW distance 2.70.
Read it as	Every number carries a range. A −3 mV peak shift is inside the ±5 mV normal band, so no lithium-loss flag.
Decide	Quote the range, not the single number.

What it cannot tell you. The exact ageing mechanism inside a cell. That needs lab tests. The tool tells you which cell to test.

CHAPTER 4

How long will it last, and how sure are you?

Remaining life as a range, the knee, and why one day of data can say anything.

Who asks: asset managers, CFOs, lenders, lessors.

QUICK ANSWER

The remaining-life tile: years and cycles to 80% health, with a likely range, and which cell limits it. The outlook chart shows the path.

Where in the app: Quick page → *See detailed analytics* → **Life forecast** (MW walk, end-of-life date, "Why early or late?"). Health curve also on **Pack overview**.

The knee: the plot twist that costs money

A battery fades slowly for a long time, then faster. The bend is the knee. Surprise replacements, warranty claims and second-life values all hang on when it comes.

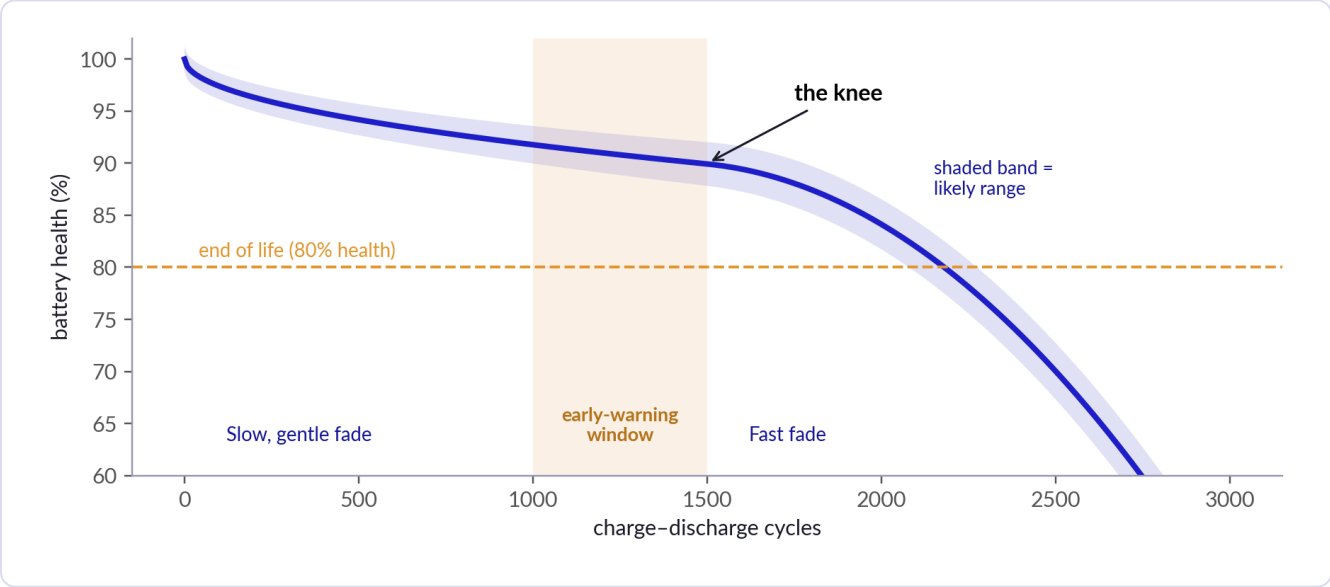


Figure 4.1 — Illustration. Slow fade, an early-warning window, then fast fade. The shaded band is the likely range.

- **Before the knee:** slow film growth and a little lithium lost. A hundred cycles barely move health.
- **After the knee:** something new starts — usually lithium plating or particle cracking. The same hundred cycles cost far more.

A simple curve-fit sees the knee only after it has happened. The app reports a knee only when the data clearly bends. Rising resistance and slower settling often show up before capacity drops — so a repeat log is the cheapest early warning.

How can one day of data say anything?

Think of a doctor. Today's blood pressure does not tell your whole future. Combined with what the doctor knows about people like you, it places you on a familiar path. The engine already knows how LFP cells age. Your log is the position fix that places this cell on that path.

Question	From one 12–24 h log?	What the app does
How healthy is it today?	Yes	Every forecast starts from measured health.
How resistant is each cell?	Yes	Corrected to 25 °C, with a range.
How is it used?	Yes	The log's duty becomes the forecast's default.
How fast will this cell fade?	Not from one day	Uses typical LFP behaviour, adjusted by the cell's resistance. A second log sharpens it.
When will the knee come?	Not exactly	Gives a hint and a range. A repeat log helps most.

The full forecast (Detailed)



You see	Today's health (dot), the forecast and its range (blue), the 80% line. Demo pack: end of life at about 1,620 cycles (likely 1,470–1,727); a little under five years at 0.91 cycles a day.
Read it as	The band widens further out and narrows as you add logs. It covers only the model's own doubt, not label or sensor errors.
Decide	Replacement timing and budget. Plan for the bad end of the range, not the middle.

Situation	Action
Knee possible, resistance rising	Watch list. Upload a fresh log every few weeks.
Hot, or fast-charged in the cold	Change the habit first (Chapter 9).
Knee likely inside warranty term	Review warranty reserve now (Chapter 5).
No knee reported	Good, not a promise. Re-run with the next log.

What it cannot tell you. An exact knee date from one day of data. Ranges narrow only as logs accumulate.

CHAPTER 5

What warranty can I offer, and what reserve should I hold?

Turn a forecast range into a probability you can price.

Who asks: OEMs, pack makers, importers, financiers, insurers, second-life sellers.

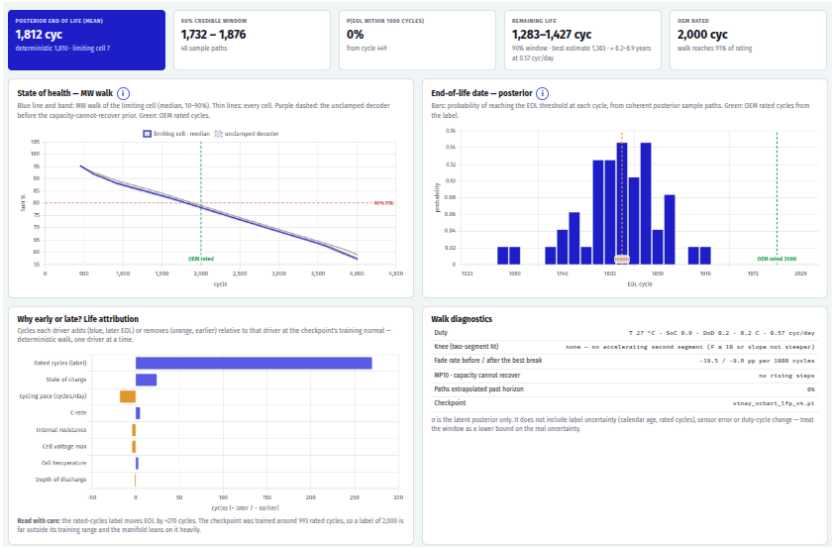
QUICK ANSWER

The life and usage tiles tell you whether the pack reaches 80% before or after its rated cycles at today's duty (worked example in Chapter 2). Detailed mode gives the probability of lasting N cycles.

Where in the app: Quick page → *See detailed analytics* → **Life forecast** (end-of-life date vs rated cycles) · **Physics Lab** ("Your prediction") · **Capacity & usage** (duty).

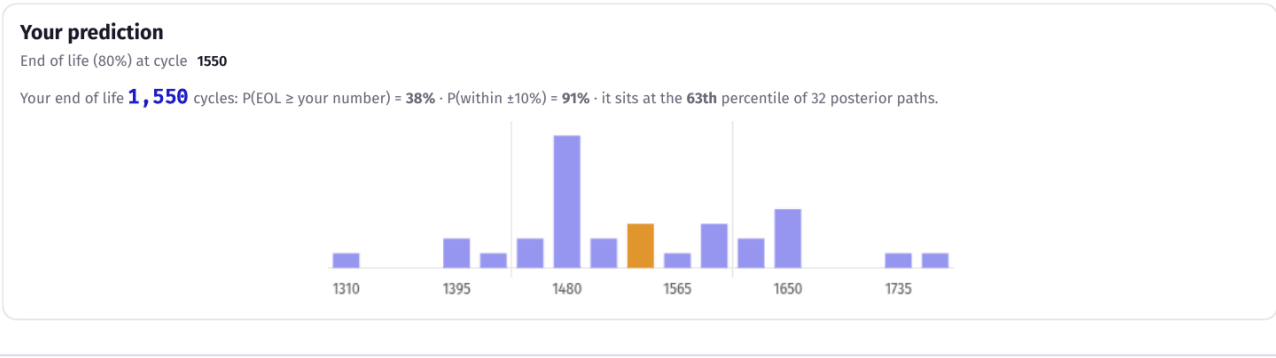
A warranty is a bet on a number of cycles or years. Most tools give you one number to bet on. A single number cannot tell you how often you will lose. A range with probabilities can.

The Quick pack's warranty, in the Life forecast tab



You see	The pack from Chapter 2. End of life at 1,812 cycles on average (90% window 1,732–1,876). Green line: the 2,000-cycle rating. The pack reaches 91% of its rating.
Read it as	The whole likely range sits below the rating. A "2,000 cycles to 80%" warranty would likely be claimed. The chance of reaching end of life within 1,000 cycles is 0%.
Decide	Offer a cycle warranty below about 1,700, or a years-based term. "Why early or late?" shows the rated-cycles label moves the answer most — confirm it from the nameplate first.

"Will it last N cycles?" as a probability



You see	Type a number, here 1,550 cycles. The bars show how the forecast paths spread. The readout: 38% chance it lasts at least that long; 91% chance it ends within ±10% of your number.
Read it as	"Will it last N cycles?" becomes a number you can use in a contract.
Decide	Set the warranty term where the chance of failing inside it is below your target. Set the reserve from that chance. Re-run every quarter.

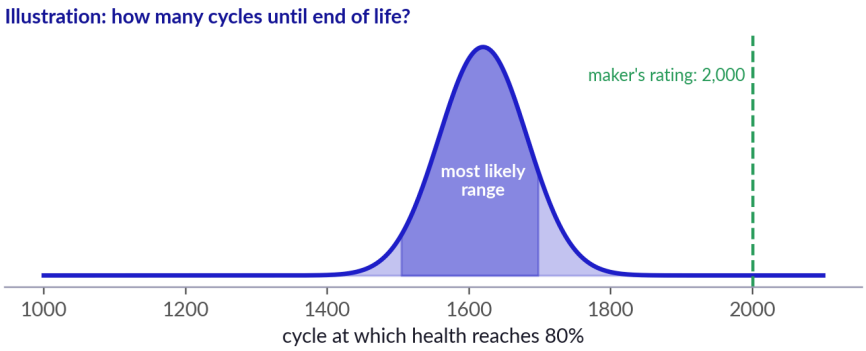
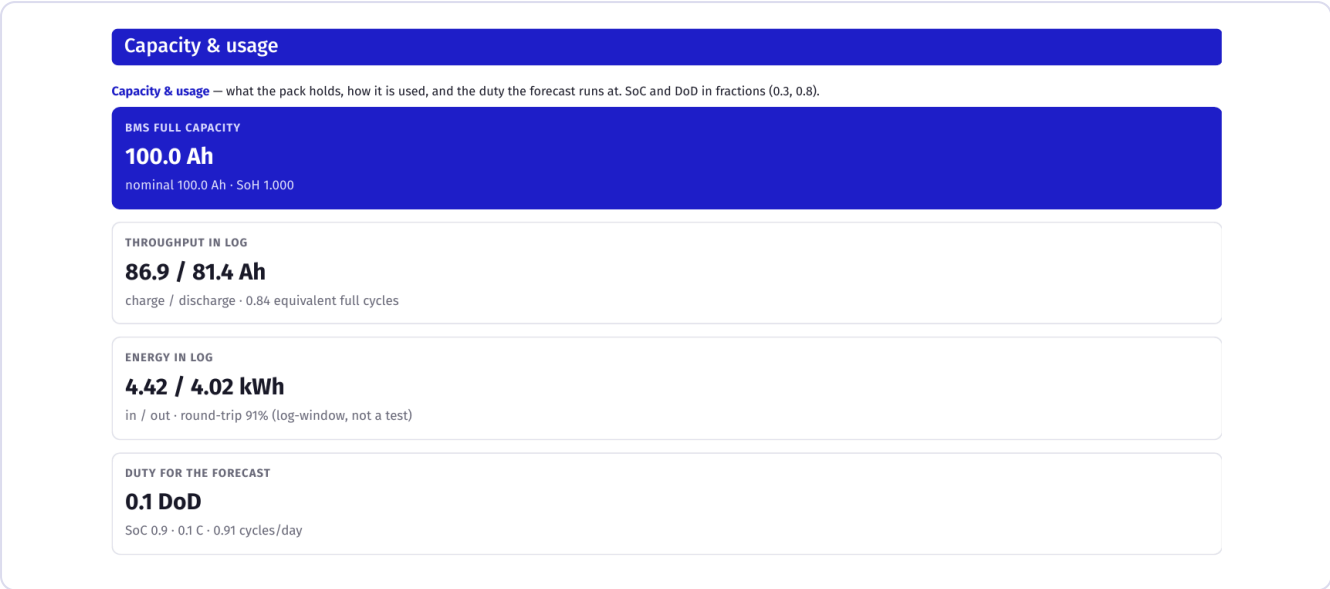


Figure 5.1 — Illustration. The likely end-of-life range against a maker's rating of 2,000 cycles. When the rating sits outside the range, the warranty has a problem.

What the pack has actually done



You see	BMS capacity (100 Ah), throughput in the log (86.9 Ah in / 81.4 Ah out, 0.84 equivalent full cycles), energy and round-trip efficiency, and the duty the forecast assumes (0.1 depth, 0.9 charge level, 0.1 C, 0.91 cycles a day).
Read it as	The forecast is only valid at the stated duty. Cycles a day turns cycles into years.
Decide	Write the duty into the warranty terms. A customer who doubles the duty halves the calendar life.

A warranty checklist

Check	Where
Real age and rated cycles entered?	Pack label (Ch 3). Defaults make the answer worthless.
Rated cycles inside the likely end-of-life range?	Life forecast (Ch 4).
Chance of failing inside the term below target?	"Your prediction" above.
Is the duty in the log the duty in the contract?	Capacity & usage (Ch 9).
Is the BMS health number trustworthy?	Check MP10 (Ch 11).
Is the pack outside what the engine knows?	MW distance (Ch 12). If yes, do not price on it.

What it cannot tell you. The forecast range is currently too tight: in our public-data test it caught the true answer 12–25% of the time, against a target of 90%. Treat it as the minimum uncertainty and add your own margin (Chapter 17).

CHAPTER 6

Which cell is the problem, and do I send a technician?

The pack map, eight views of the same cells, and the per-cell table.

Who asks: maintenance leads, service partners, warranty engineers.

QUICK ANSWER

The cell grid on the Quick page marks cells "normal", "check" or "no signal". Detailed mode tells you *why* a cell is flagged, and whether the difference is real.

Where in the app: Quick page → *See detailed analytics* → **Pack overview** (pack map, eight layers) · **Cell internals** (resistance ranges, per-cell table).

The pack map



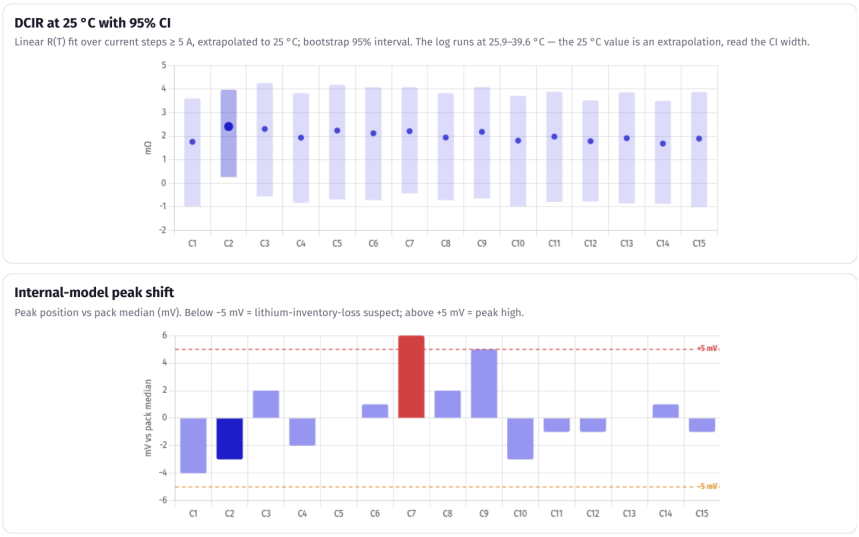
You see	Sixteen cells as wired, coloured by internal resistance at 25 °C. Cell 16 is hatched: its sense wire is dead. Tags mark DEAD and PEAK HIGH cells.
Read it as	Colours are relative to the pack. Click a cell and every screen refocuses on it.
Decide	A dead channel is a wiring job. Position tells the technician where to walk.

Eight views of the same pack



You see	The same 16 tiles recoloured: engine health, resistance, charging-peak shift, settling time, MW distance, resting voltage vs pack, temperature and checks failed.
Read it as	A cell that stands out on *more than one* view is the real suspect. One odd view is a note, not an alarm. Read the numbers, not just the colours: sixteen near-identical cells still show a "worst" one.
Decide	Capacity worry → health and peak shift. Power or heat worry → resistance and settling time.

Is the difference real?



You see	Top: each cell's resistance with its likely range. Bottom: charging-peak shift vs the pack median, with ±5 mV lines.
Read it as	If two ranges overlap, the cells are not really different. Here every resistance range overlaps: no resistance outlier.
Decide	Overlapping ranges → no truck roll. That saves money too.

The per-cell table: the audit trail

Per-cell table

Click a row to focus the cell everywhere. Two resistance figures are shown on purpose: the internal-model R25 (T-normalised, CI) and the 16-parameter converter's median |ΔV/ΔI| at log temperature — they are different estimators.

Cell	Label	R25 mΩ	95% CI	dR/dT	16p IR mΩ	Peak V	Shift mV	Width mV	Height	τ s	V mean	V range mV	Steps
1	NOMINAL	1.76	-1.0–3.6	+0.141	2.540	3.383	-4.0	35	537.15	76	3.392	355	28
2	NOMINAL	2.41	0.3–4.0	+0.077	2.213	3.384	-3.0	65	418.57	73	3.383	312	29
3	NOMINAL	2.31	-0.5–4.2	+0.140	2.484	3.389	+2.0	65	447.56	78	3.392	361	28
4	NOMINAL	1.94	-0.8–3.8	+0.141	2.453	3.385	-2.0	35	516.45	79	3.395	351	28
5	NOMINAL	2.24	-0.7–4.2	+0.153	2.424	3.387	0.0	65	418.13	71	3.390	370	28
6	NOMINAL	2.12	-0.7–4.1	+0.142	2.366	3.388	+1.0	35	473.84	66	3.393	357	28
7	PEAK HIGH	2.21	-0.4–4.1	+0.135	2.273	3.393	+6.0	65	455.29	71	3.393	343	28
8	NOMINAL	1.95	-0.7–3.8	+0.140	2.414	3.389	+2.0	60	481.80	71	3.395	353	28
9	NOMINAL	2.19	-0.6–4.1	+0.137	2.354	3.392	+5.0	60	456.05	74	3.395	355	28
10	NOMINAL	1.81	-1.0–3.7	+0.141	2.540	3.384	-3.0	60	471.42	69	3.393	356	28
11	NOMINAL	1.98	-0.8–3.9	+0.136	2.533	3.386	-1.0	60	469.14	72	3.392	350	28
12	NOMINAL	1.79	-0.8–3.5	+0.125	2.305	3.386	-1.0	60	493.52	73	3.393	342	28
13	NOMINAL	1.92	-0.8–3.9	+0.138	2.550	3.387	0.0	60	488.41	70	3.394	347	28
14	NOMINAL	1.69	-0.9–3.5	+0.136	2.410	3.388	+1.0	35	566.11	73	3.396	347	28
15	NOMINAL	1.90	-1.0–3.9	+0.157	2.449	3.386	-1.0	40	522.48	75	3.394	356	28
16	DEAD	–	–	–	0.000	–	–	–	–	–	–	–	–

You see	Resistance and range, resistance vs temperature, peak position, shift, width and height, settling time, voltages and a one-word label: NOMINAL, PEAK HIGH, DEAD, LLI SUSPECT or high_IR.
Decide	Scan the label column first. Anyone can recompute a row from the raw CSV.

Column	What it tells you
Resistance (25 °C)	Rising = power loss and heat. Well above peers = high_IR candidate.
Resistance vs temperature	Must be negative. Positive = bad connection or sensor.
Settling time	Longer = slower movement inside the cell.
Peak shift	More than 5 mV below the pack = possible lithium loss.
Peak width / height	Wider = uneven electrode. Shorter = lost active material.
Measurement floor	Differences below it are noise (Chapter 12).

What it cannot tell you. Whether a flagged cell is safe. A flag says "look here", not "this is dangerous".

CHAPTER 7

Weak cell or balancing job? How much energy am I losing?

Balance decides usable capacity. A cheap fix often hides behind an expensive-looking symptom.

Who asks: operations, technicians, customers complaining of short range or runtime.

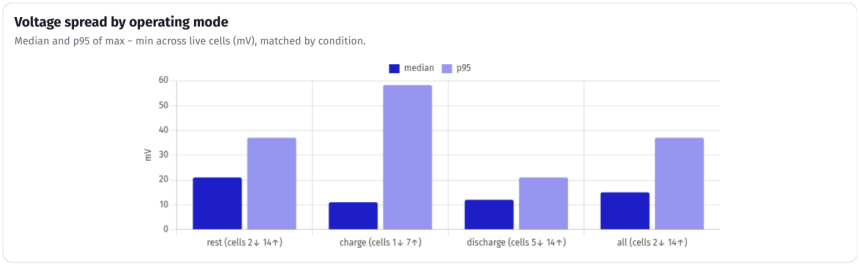
QUICK ANSWER

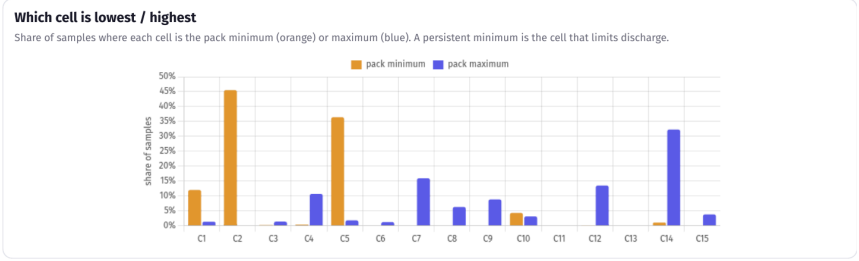
The usable-energy tile shows energy today vs new. "Cells out of balance" in What to do gives the resting spread (533 mV on the Chapter 2 pack). Often the fix is a full charge so the BMS can balance.

Where in the app: Quick page → *See detailed analytics* → **Thermal & balance** (spread by mode, lowest / highest cell) · **Telemetry replay** (cell voltages).

In a series pack, the lowest cell ends every discharge and the highest cell ends every charge. One cell out of line costs the whole pack energy, even if every cell is healthy. Balancing is cheap. Replacing a cell is not. Know which one you need.

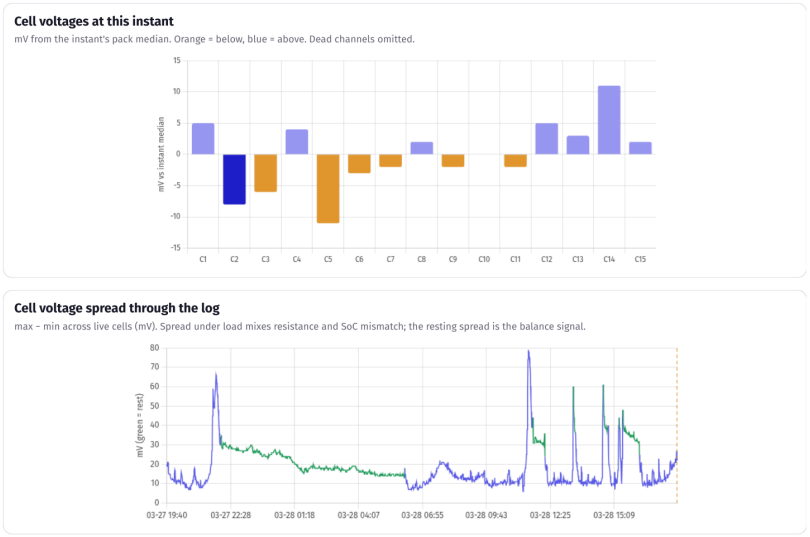
Which cell sets the limit?





You see	Top: gap between highest and lowest cell at rest, charge and discharge. Bottom: how often each cell is the pack's lowest (orange) or highest (blue).
Read it as	The spread **at rest** is the balance signal. A cell that is persistently lowest sets usable capacity.
Decide	Lowest more than half the time → balance it, then include it in the next capacity test.

Through the log



You see	Each cell vs the pack median at one moment, and the spread through the whole log.
Read it as	Spread under load mixes resistance and balance. Spread at rest is balance alone.
Decide	Resting spread above normal for your pack → balancing job. If it returns after balancing → that cell is weak.

What you see	Likely cause and action
Large resting spread, health similar across cells	Balance problem. Full charge, let the BMS balance, re-log.
Spread returns soon after balancing	Weak or self-discharging cell. Capacity test that cell.
One cell low at rest and peak shifted below −5 mV	Lithium loss. Watch list, plan a spare (Chapter 10).
One cell high on resistance, ranges not overlapping	Check its connections first, then the cell.

CHAPTER 8

Is it running hot, and where?

Heat is the biggest ageing accelerator. It is also the cheapest to find.

Who asks: maintenance, safety officers, site operators.

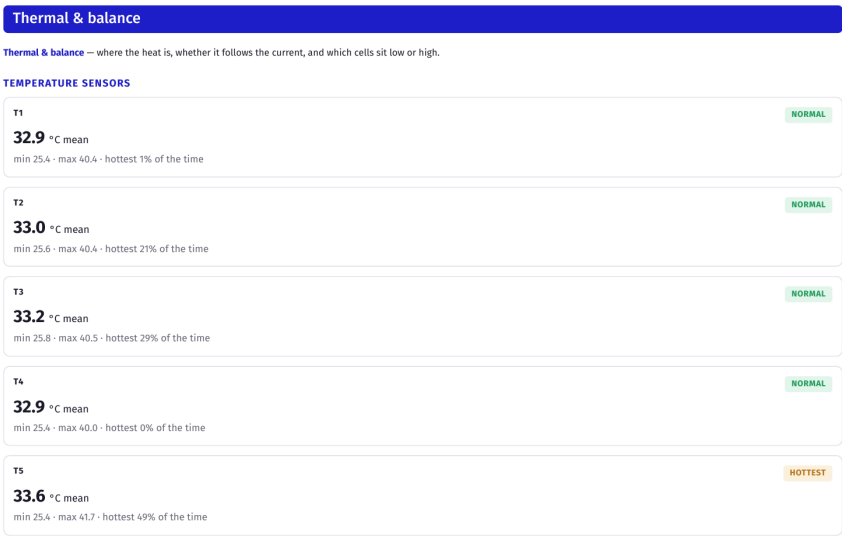
QUICK ANSWER

The temperature line on the Quick page shows average and peak temperature. Detailed mode finds the hot spot and tells you whether the heat is normal (follows current) or local (a joint, busbar or airflow).

Where in the app: Quick page → *See detailed analytics* → **Thermal & balance** (sensors, Joule check) · **Telemetry replay** (temperatures).

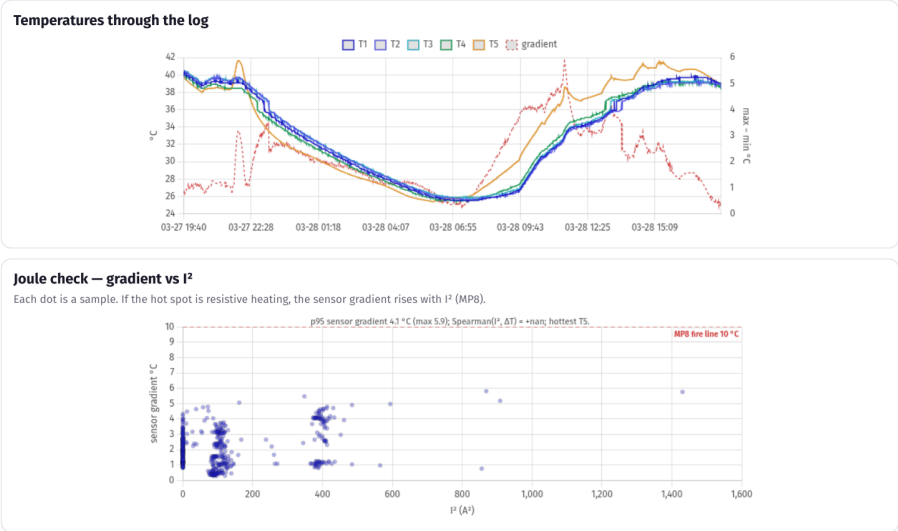
At the heat sensitivity the engine learned, a cell at 39.5 °C ages about 2.5 times faster than at 25 °C. In an earlier reference pack, one hot spot already cost about 100 cycles.

Which sensor is hottest, and how often



You see	One card per sensor: average, lowest, highest and share of time it is the hottest. Demo pack: T5 is hottest 49% of the time.
Read it as	A sensor that is hottest half the time is a *place*, not a coincidence.
Decide	Inspect near that sensor: torque check, thermal camera, cooling path.

Normal heat or a local fault?



You see	Top: all sensors over time. Bottom: sensor-to-sensor gap against current squared, with a 10 °C alarm line.
Read it as	Ordinary resistive heat grows with current squared. Spikes at low current mean the heat is local: a loose joint, a busbar, poor airflow or a faulty sensor.
Decide	Tight link to current → cooling capacity. Loose link → send a technician to that spot.

What it cannot tell you. Thermal runaway risk. That needs a real-time safety system. This tool reads a log after the fact.

CHAPTER 9

Which operating habit should I change, and does it pay?

Test-drive a setpoint change in software before you touch the real pack.

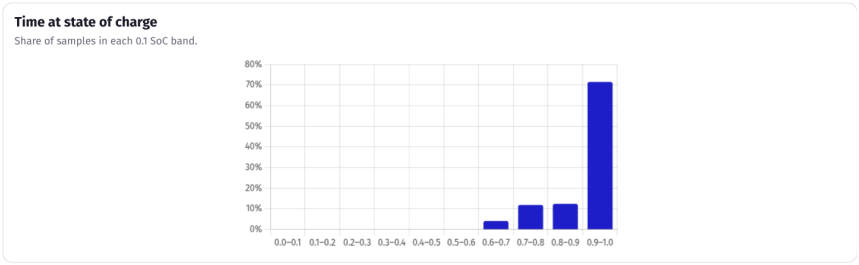
Who asks: operations heads, fleet managers, BESS dispatch teams.

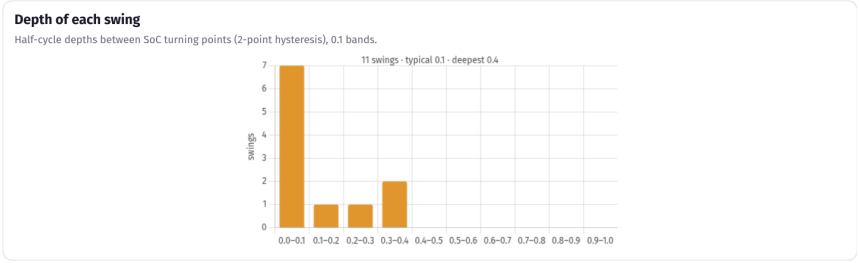
QUICK ANSWER

Not on the Quick page. Use Detailed: Capacity & usage shows how the pack is used; the Physics Lab and Virtual Lab price each change in cycles and years.

Where in the app: Quick page → *See detailed analytics* → **Capacity & usage** · **Physics Lab** · **Virtual Lab**.

How is the pack used today?



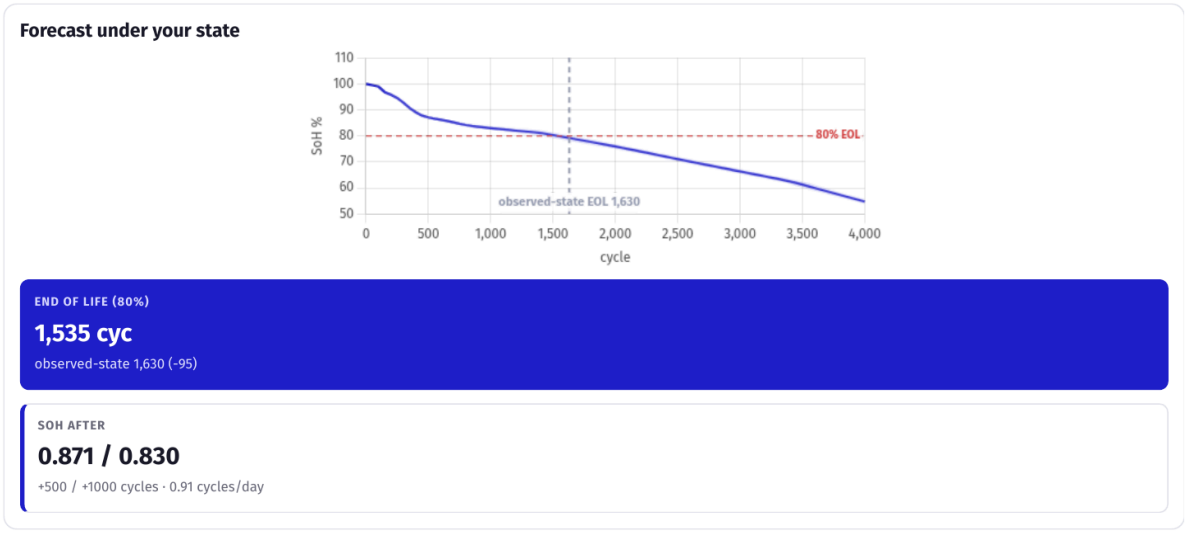


You see	Time in each charge band (the demo pack spends most of its time above 90%) and how deep each swing goes.
Read it as	Sitting near full is a storage-ageing habit. Shallow swings are gentle.
Decide	**The cheapest lever:** cap the resting charge level. Test it in the Lab first.



Figure 9.1 — Charge rates and charge accounting. A fast-charge-heavy fleet should watch the plating check.

The Physics Lab: one what-if at a time



You see	The forecast for a state you set with sliders. Here end of life at 1,535 cycles vs 1,630 for the observed state: −95 cycles.
Read it as	The gap between the two forecasts is the cost or gain of your change.
Decide	Compare two what-ifs side by side before changing anything in the field.

The Virtual Lab: rank every lever

Tick the habits to vary: charge rate, cycling pace, depth of discharge, resting charge level, temperature. Everything else stays at the logged duty.

Output	What it means	Decision
Lever ranking	Effect of each habit on end of life, in cycles and years	Spend on the biggest lever
Response curve	Effect against the setting, e.g. 15 → 45 °C	Find where more cooling stops paying
Verdict	A habit is "resolved" only if its effect beats the forecast range	"NOT resolved" → do not spend on it

Why this matters. Most what-if tools give a number for every change. This one also tells you when the number is within the noise. "NOT resolved" is often the most valuable sentence in the app: it stops spending that would not show up in battery life. The Virtual Lab has reproduced the published findings of leading peer-reviewed battery researchers on their own experimental data.

CHAPTER 10

What is this used pack worth? Resale, second life, claims

Grade cells before you buy, sell, refurbish or pay a claim.

Who asks: used-pack buyers and sellers, refurbishers, second-life integrators, insurers, lenders taking a pack as collateral.

QUICK ANSWER

Health (weakest cell), remaining life and cycles used against rating. Together they give a first price band. Detailed mode grades each cell and checks the BMS health number.

Where in the app: Quick page → *See detailed analytics* → **Cell internals** (peak shift, signature, settling time) · **Pack overview**.

A used pack is priced on three things: how much capacity is left, how long it will keep it, and whether the seller's numbers are honest. The app answers all three from one log.

The earliest capacity warning: the charging peak

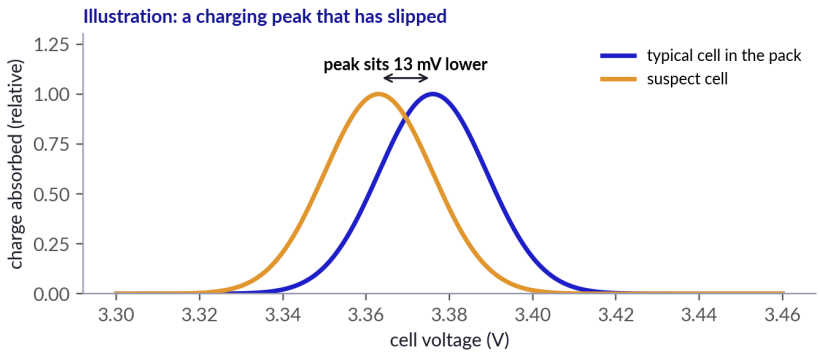
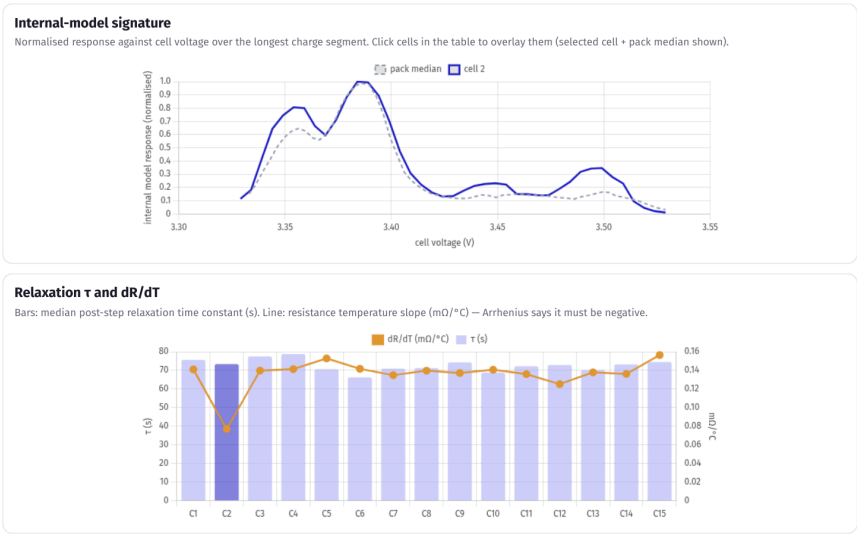


Figure 10.1 — Illustration. Where the peak sits tracks lithium loss. Its width tracks how even the electrode is. Its height tracks how much material still works.

LFP's flat voltage hides ageing from the BMS. The charging peak is one of the few signals that moves early, and the app pulls it from ordinary CAN data without a lab test.

Peak shape and settling time



You see	Top: the charging response of one cell against the pack median. Bottom: settling time per cell, and how resistance changes with temperature.
Read it as	Different peak shapes mean different electrode behaviour. Longer settling means power ability is falling before capacity is.
Decide	Grade cells: same peak and settling as the pack → grade A. Shifted peak or slow settling → grade down or swap.

Worked example (earlier reference pack). One cell's charging peak sat 13 mV below the pack median. Its health matched its neighbours and the BMS read 100%. The signal showed up roughly 1,600 forecast cycles before the 80% line. For a buyer, that is one cell to disclose or swap. For a seller, it is one targeted capacity test instead of sixteen. Trend across logs (−13 → −20 mV) confirms active lithium loss. It does not justify replacing the cell today.

Valuation question	Where to look
Capacity left?	Health tile; per-cell health (Ch 3)
Life left at the buyer's duty?	Physics Lab with the new duty (Ch 9)
Is the seller's SoH honest?	Engine vs BMS health; MP5 and MP10 (Ch 11)
Which cells to disclose or swap?	Peak shift, per-cell labels (Ch 6)
Was it abused?	Temperature history, C-rate, time at full (Ch 8, 9)

CHAPTER 11

Is the data honest? Sensor fault, wrong setting or tampering

CAN has no passwords. Physics is hard to fake.

Who asks: security teams, auditors, insurers, buyers of used packs, anyone settling a dispute.

QUICK ANSWER

"Health checks flagged" on the Quick page counts how many of ten physics checks need a look. Detailed mode shows which check, on which cell, and what to suspect.

Where in the app: Quick page → *See detailed analytics* → **Pack overview** (ten checks) · **MW manifold & physics** (checks across the pack) · **Data & conversion** (coverage) · **Telemetry replay**.

This chapter covers everyday data problems. For deliberate, Stuxnet-style attacks, see Chapters 14 and 15.

Any device on a CAN bus can send a believable message. But a made-up charge level still has to match the current that flowed. Fake voltages still have to move with current. The ten checks test exactly that. A failed check says *something is inconsistent*. A human decides: sensor, setting or attack.

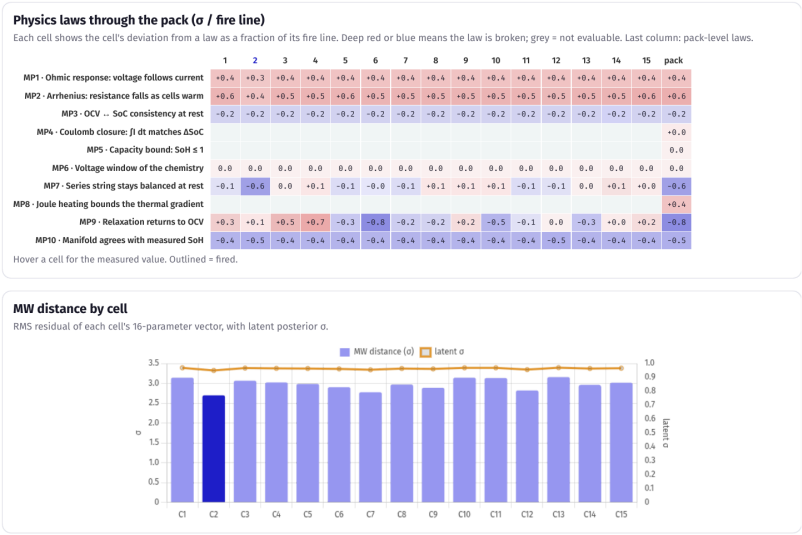


You see	Ten bars, each against its alarm line. Red means failed. Tags say *physical* (a law of physics) or *empirical* (a pattern of healthy cells).
Decide	A failed check is a question for a human before it is a number in a report.

Check	Asks whether…	If it fails, suspect…
MP1	voltage moves with current (Ohm's law)	timing mismatch, wrong channel, faked voltage
MP2	resistance falls as cells warm	bad joint, misplaced sensor

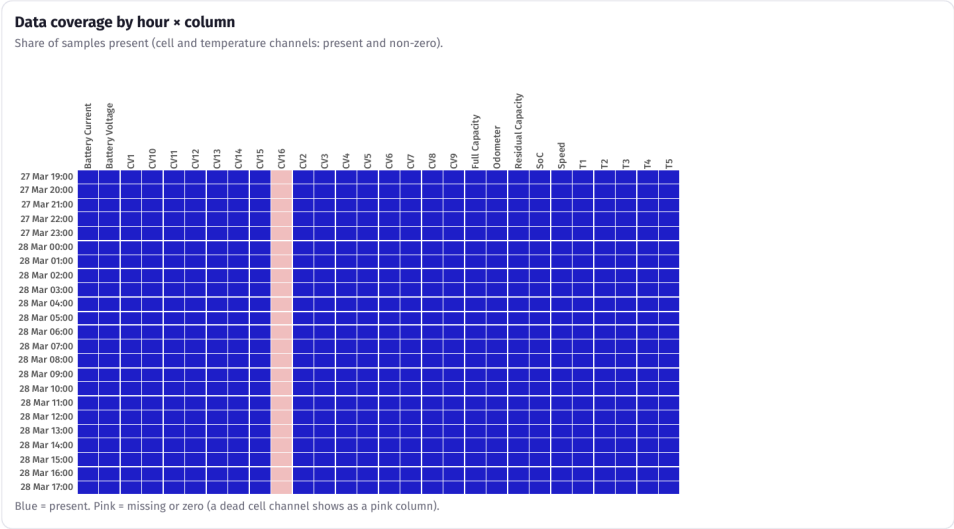
MP3	resting voltage agrees with BMS charge level	charge estimate drifting or faked
MP4	current in and out matches the charge change	current sensor error, falsified charge level
MP5	capacity is not above nameplate	reset or doctored health value
MP6	voltages stay in the LFP window (2.50–3.65 V)	sensor fault, abuse, injected values
MP7	cells are balanced at rest	weak or self-discharging cell
MP8	heat spread matches current (I^2 heating)	hot joint, cooling fault, faked sensor
MP9	voltage settles back after a load step	slow internals, replayed or flat-lined voltage
MP10	engine's health agrees with the BMS	BMS health drift or manipulation

Across the whole pack



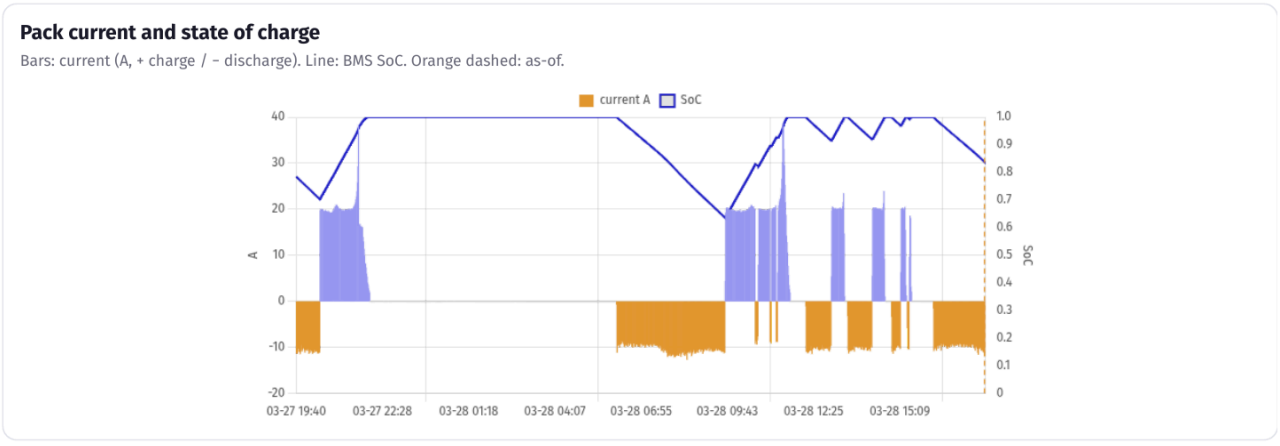
You see	Each cell's result on each check as a fraction of the alarm line, and how unusual each cell looks.
Read it as	Strain on every cell points to the pack or the log. Strain on one cell points to that cell.
Decide	Use the single pass-score to screen many packs; the cell column shows where to look.

Is any channel missing?



You see	How much of each column is present each hour. A pink column is a dead channel (here CV16). A pink row is a logger gap.
Decide	Fix the data path before you doubt the cell. A BMS blind to one cell is a safety gap in its own right.

Replay the moment



You see	Current and BMS charge level through the log. The Telemetry replay slider shows every reading at any moment.
Decide	When a finding looks odd, slide to the moment it happened. It turns "is that real?" into a look.

CHAPTER 12

Can I trust this number?

The screens that tell you when the engine is out of its depth.

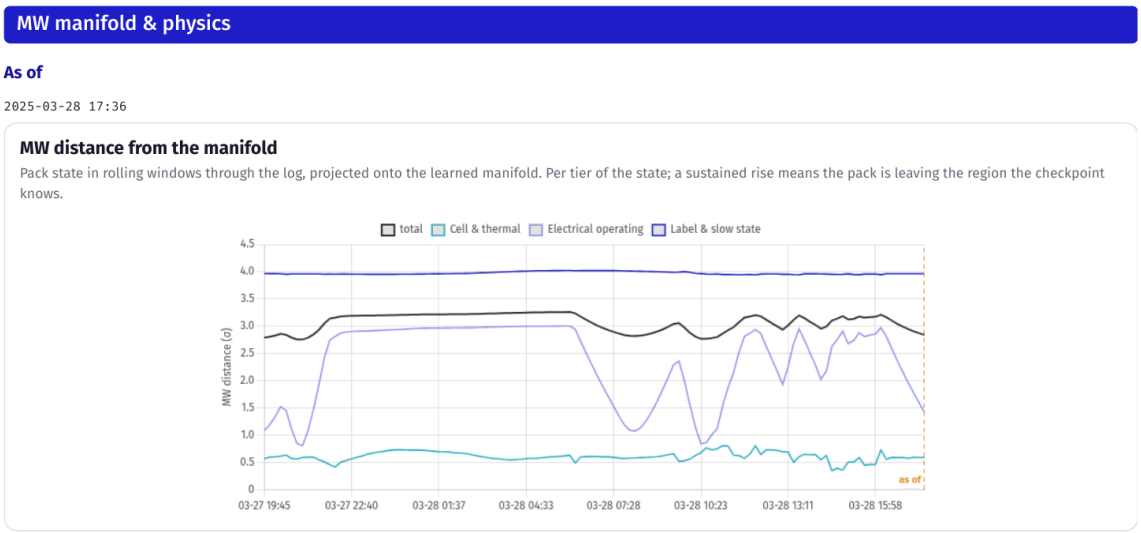
Who asks: anyone about to quote a number in a report, contract, claim or board paper.

QUICK ANSWER

Every Quick tile carries a range. "Battery age assumed" warns when the label is a default. For a contract, check the four screens below first.

Where in the app: Quick page → *See detailed analytics* → **MW manifold & physics** (MW distance, residuals, training range) · **Data & conversion** (measurement floor).

1 · Is the pack familiar to the engine?

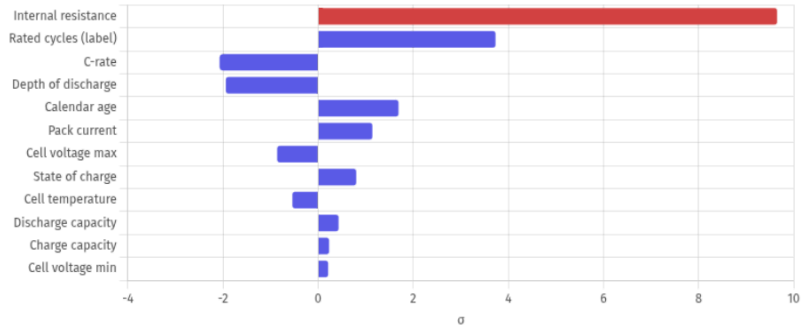


You see	How unusual the pack's state looks through the log ("MW distance"), split into groups.
Read it as	Flat = familiar territory. A steady rise = conditions the engine has not seen. Above about 3 is unfamiliar ground.
Decide	Rising line → do not rely on the forecast; gather more data.

2 · Which input is the problem?

Off-manifold residual by feature

Standardised residual (σ) after projecting the as-of window onto the manifold. Red = the feature the manifold disagrees with most.



You see How far each input is from what the engine expects. Biggest in red: internal resistance, then the rated-cycles label.

Decide Fix the inputs you control first (the label). Then decide if what remains is a cell issue or a pack unlike the training batteries.

3 · Is it inside the training range?

Pack vs the checkpoint's training range

z of the pack median against the checkpoint normaliser. |z| > 3 means the manifold is extrapolating on that input.

Input	Pack	Training mean ± sd	z
Pack current	0.384 A	-3.644 ± 41.016	+0.1
Mean cell voltage	3.393 V	3.343 ± 0.154	+0.3
Cell temperature	33.020 °C	28.098 ± 7.529	+0.7
Cell voltage min	3.251 V	3.296 ± 0.164	-0.3
State of charge	0.851	0.635 ± 0.184	+1.2
Cycle count	0.000 cyc	322.692 ± 245.373	-1.3
Internal resistance	2.424 mΩ	0.854 ± 0.170	+9.3
Charge capacity	98.000 Ah	94.658 ± 10.484	+0.3
Discharge capacity	100.000 Ah	94.660 ± 10.482	+0.5
C-rate	0.004 C	0.361 ± 0.197	1.8

You see The pack's typical value for each input against the training batteries. Internal resistance is far outside (+9.3).

Decide Any input beyond about 3 goes into the caveats of your report.

4 · Can the sensors even see the difference?

Measurement floor

Rows · sample interval	1351 · 60 s
Span	2025-03-27 19:40 → 2025-03-28 17:50 (22.2 h)
Current / voltage resolution	0.01 A / 1 mV
Typical current step	14.3 A
IR measurement floor	0.070 mΩ
Checkpoint	vinay_vchart_lfp_v4.pt (V5_ULTIMATE)
Label inputs	500 d (default) · 2000 rated (default) · LFP · via CAN converter

You see	Rows, interval, time span, current and voltage resolution, typical current step, and the resistance floor (0.070 mΩ).
Read it as	Differences below the floor are noise. Some things no log can supply: a calibration chain, a test protocol, a day-one baseline.
Decide	Attach this table to any finding. It answers an auditor before they ask.

> **The trust test in four lines.** Real label entered. MW distance below about 3. No input far outside training. Differences above the floor. If all four hold, quote the range. If not, say which one failed.

CHAPTER 13

How do I run this across a fleet without an IT project?

Any log format, up to 100 at once, no integration.

Who asks: fleet owners, swapping operators, OEM service heads, BESS asset managers.

QUICK ANSWER

Drop up to 100 logs. Each is analysed on its own and listed on a Fleet screen. A small labels.csv sets age and rated cycles pack by pack.

Where in the app: Quick page → *See detailed analytics* → **Data & conversion** (converter, fleet upload) · **Fleet**.

Step	What happens
1 · Upload	A BMS / CAN CSV. The standard layout has 29 columns (current, pack voltage, cell voltages CV1–CV16, capacity, charge level, speed, odometer, temperatures T1–T5, timestamp).
2 · Convert	Any other layout goes through the converter. An AI reads only the column names and writes a mapping (names, units, sign, timestamp). The app applies the mapping itself and checks it. The AI never touches your data rows.
3 · Label	Enter age, rated cycles, capacity and chemistry, or upload labels.csv for a fleet.
4 · Quick for all	Health, life, energy, use and a to-do list for every pack in seconds.
5 · Detailed for some	Tap See detailed analytics on the packs that matter: warranty claims, resale, incidents.
6 · Repeat	Upload a new log every few weeks. Ranges narrow and trends appear.

A fleet routine that works

- Every month: Quick on every pack. Sort by verdict, then by remaining life.
- Every "Needs attention" pack: hand What to do to the technician.
- Bottom 10% on remaining life: Detailed, then Chapter 16's playbook.
- Every quarter: re-run warranty probabilities (Chapter 5).

PART 4

Battery cybersecurity

The most dangerous attack on a battery is not the one that stops it. It is the one that damages it while every screen says "normal". This part explains that attack, why network tools miss it, and how physics catches it.

CHAPTER 14

Stuxnet-style attacks on batteries, and why network monitors miss them

When the data lies, the network looks perfectly normal.

Who asks: BESS owners, utilities, EV and swapping fleet operators, insurers, security heads, regulators.

QUICK ANSWER

A Stuxnet-style attack makes a battery do harm while its data says all is well. The traffic is legal, so network tools see nothing wrong. The numbers, however, must still obey physics. Checking that is what Xtreme Analytics does.

What Stuxnet taught us

Stuxnet, discovered in 2010, attacked the controllers running uranium centrifuges. It changed their speeds to wear them out. At the same time it replayed recorded "normal" readings to the operators. The screens looked healthy while the machines were being damaged.

The lesson is simple. A sophisticated attacker does not just break the process. They **hide** the damage behind believable data. Security that trusts the data is blind to that attack.

The same attack on a battery

Battery data travels on CAN, from cells to BMS to EMS or vehicle. CAN has no passwords and no sender check. Any device on the bus, or a compromised BMS or EMS, can send a frame that looks perfectly valid.

Attack	What the attacker does	Business damage
Masked overcharge	Pushes cells past their safe limit, while CAN reports normal voltages.	Lithium plating, heat, fire risk. Loss of the asset.
Fake temperature	Reports cool cells while they run hot. Cooling never starts.	Fast ageing; thermal risk.
Doctored charge level	Falsifies SoC so the pack is driven too full or too empty.	Hidden damage; stranded energy; trading losses for BESS.
Doctored health	Inflates SoH or resets the counter.	Resale, warranty and insurance fraud.
Replay	Plays back a recorded healthy log to hide what is really happening.	Every downstream decision is made on fiction.
Slow sabotage	Nudges setpoints: higher resting charge, faster charging, warmer operation. Nothing alarms.	Pack ages years early. Looks like bad cells, not an attack.

Blinding	Kills or freezes a cell's sense channel.	The BMS cannot protect a cell it cannot see.

Note the last two. Slow sabotage and blinding cause no outage and no obvious incident. They show up as an "early failure" or a "warranty dispute" — and are written off as bad luck.

Three layers of security, and what each one sees

IT security protects computers and accounts. OT security protects industrial networks. Most OT security products are, at heart, **network monitors**: they build an asset inventory, decode industrial protocols and flag unusual traffic, often with an intrusion-detection engine such as Suricata underneath. That work matters, and every serious site needs it.

But in a Stuxnet-style attack the traffic is not unusual. The right device sends the right command in the right format. Only the *values* are false. A network monitor asks "who talked to whom, and how?" It does not ask "can these numbers be true together?"

Layer	Watches	Catches	Misses
IT security	Servers, accounts, email, endpoints	Malware, stolen passwords, ransomware	Anything on the battery's own CAN bus
OT network security	Industrial network traffic, assets, protocols	New devices, odd commands, known attack signatures	Legitimate-looking frames carrying false values; CAN traffic inside the pack that never reaches the plant network
Physics-level (Xtreme Analytics)	The battery's measured values	Data that breaks the laws of physics or the behaviour of real LFP cells	Who did it and how they got in. That is the network layer's job.

> **The point.** These layers do not compete. Network security guards the door. Physics checks whether what came through the door is telling the truth. A Stuxnet-style attack walks through the door with valid credentials. Only the second check is left.

Why physics is hard to fake

Faking one reading is easy. Faking a whole battery consistently is hard. A believable fake must keep, at every second, for every cell:

- voltage moving with current exactly as the cell's resistance dictates;
- charge level matching the current that actually flowed in and out;
- resting voltage matching the charge level on the LFP curve;
- heat rising with current squared, and resistance falling as cells warm;
- voltage settling after each load step the way real LFP chemistry does;
- health that only ever goes down.

An attacker who controls the data must model all of this, for every cell, in real time. Most will not. That gap is where detection lives.

What this chapter does not claim. A well-resourced attacker with a full battery model could forge physically consistent data. Physics checks raise the cost of an attack sharply. They do not make it impossible.

CHAPTER 15

How Xtreme Analytics catches them: physics as the last line of defence

Each attack leaves a physical fingerprint. Each fingerprint has a screen.

Who asks: security teams, BESS operators, incident responders, insurers investigating a loss.

QUICK ANSWER

On the Quick page, "health checks flagged" is the first alarm. In Detailed analytics, the ten physics checks, the MW manifold and the telemetry replay show which law broke, on which cell, and when.

Where in the app: Quick page → *See detailed analytics* → **Pack overview** (ten checks) · **MW manifold & physics** · **Telemetry replay** · **Capacity & usage** · **Thermal & balance** · **Data & conversion**.

Attack → fingerprint → screen

Attack	Physical fingerprint	Check	Where to look
Masked overcharge	Reported voltage does not move with current as resistance requires; values drift outside the LFP window	MP1, MP6	Pack overview; MW manifold & physics
Fake temperature	Resistance does not fall as the "temperature" rises; heat does not follow current squared	MP2, MP8	Thermal & balance (Joule check)
Doctored charge level	BMS SoC disagrees with the integrated current; resting voltage disagrees with SoC	MP3, MP4	Capacity & usage (charge accounting)
Doctored health	Capacity above nameplate; BMS SoH far from the engine's own estimate	MP5, MP10	Pack overview (two health numbers)
Replay or freeze	Voltage does not settle after load steps the way a live cell does; flat lines; repeated segments	MP9	Telemetry replay; Data & conversion (coverage)
Slow sabotage	Duty shifts: more time near full, higher charge rate, warmer cells; ageing faster than the logged duty explains	Trend	Capacity & usage; Life forecast across logs; Virtual Lab prices the damage
Blinding	A channel goes dead or frozen	Coverage	Data & conversion; pack map (hatched cell)
Any forged state	The pack's state drifts away from anything a healthy LFP pack does	MW distance	MW manifold & physics (rising line)

Two layers of physics

- **Layer 1 — is this moment physically possible?** Every window of data is checked against the ten laws and placed on the MW manifold, the engine's learned map of what healthy LFP cells can do. A forged state sits off that map.
- **Layer 2 — is this trajectory physically possible?** The engine forecasts how the pack should age at its logged duty. A pack that ages faster, or behaves differently, than its own duty explains is a question to ask — whether the cause is a fault or an attacker.

A real example from the Quick pack

On the pack in Chapter 2, two of ten checks fired. In **Capacity & usage**, the charge-accounting check (MP4) reports that the BMS charge level and the integrated current disagree, and suggests checking the current scaling. On **Pack overview**, the voltage-window check (MP6) also fired.

Most likely, this is a scaling or sensor issue, not an attack. That is exactly the point. The app does not cry "hacker". It says "these numbers cannot all be true", and a human decides: **sensor, setting or tampering?**

Incident response with a CAN log

Step	Do this	Why
1	Pull the raw BMS / CAN log from the window of interest. Keep the original file.	Evidence. The app reads a copy.
2	Run Quick. Note "health checks flagged".	Thirty-second triage.
3	See detailed analytics → MW manifold & physics. Which law, which cell?	Narrows the suspect: sensor, BMS or bus.
4	Telemetry replay to the moment a law broke.	Turns "is that real?" into a timestamp.
5	Compare with an earlier log of the same pack.	Separates a sudden change from slow drift.
6	Hand the timestamp to the network security team.	They check who sent what, at that moment.

Bayesian Cybersecurity also stages these attacks on its own cyber-physical battery test range, so the detection is tested against real attack patterns, not only theory.

What it cannot tell you. Who the attacker is, or how they got in. Whether an inconsistency is malice or a fault — only a human with context can decide. And from uploaded logs it works after the fact, not as a real-time alarm.

PART 5

Acting on it

A number that does not change a decision is decoration. This part turns signals into actions, shows the evidence behind the engine, and gets you started.

CHAPTER 16

The decision playbook

Signal → action → when. Pin this page to a wall.

Signal	What to look for	Action	When
Cell channel reads 0 V all log	Label = DEAD	Check the sense wire and BMS channel before condemning the cell	Now
A sanity check fails	Bar past alarm line	Inspect that spot; thermal camera; torque check. Ask "sensor, setting or tampering?"	Days
Hot sensor, loose link to current	Spikes at low current	Suspect a connection, busbar, airflow or the sensor	Days
Large resting spread	"Cells out of balance"	Full charge; let the BMS balance; re-log	Days
Charging peak shifted	> 5 mV below median	Watch list; confirm next log; targeted capacity test; budget a spare	Weeks
Peak shift growing across logs	e.g. −13 → −20 mV	Lithium loss confirmed: plan replacement in a service window	Next window
Resistance above peers	high_IR, ranges apart	Check connections first, then the cell	Weeks
Settling time rising across logs	Trend	Lower charge current or plan replacement	Months
Same cell lowest most of the time	> 50% of samples	Balance; include in capacity test	Weeks
Several checks fire together, with no sensor fault found	Laws broken across cells	Treat as a possible tampering incident: follow Chapter 15 incident steps; involve the network security team	Now
Engine health ≠ BMS health	MP10 fails	Audit the BMS counter; confirm with a capacity test	Before any sale or claim
MW distance above about 3	Unfamiliar territory	Do not rely on the forecast; gather more data	Before quoting life
Life range inside warranty term	Chance of failure above target	Review warranty reserve and pricing	Quarterly
Virtual Lab: habit "resolved"	Effect > range	Act on it. "NOT resolved" → do not spend	Next change
Long time at high charge	> 70% above 0.9	Try a lower resting cap in the Lab; change the setpoint if it pays	Next change

Worked example: eight findings from an earlier reference pack

A 16-cell LFP pack, 20.6 hours of CAN data. Most actionable first.

Finding	Evidence	Action
T5 hot spot	Averaged 41.5 °C, peaked 56.8 °C, hottest 62% of the time; gap up to 15.8 °C vs 10 °C line; MP8 failed	Inspect near T5 now. Already costs about 100 cycles.
Cell 16 channel dead	0 V all log	Check the sense lead. The BMS is blind to one cell.
Cell 1 lithium-loss suspect	Peak 13 mV below median	Watch list, capacity check, spare on order.
Cell 2 limits discharge	Lowest in about 56% of samples	Balance; include in capacity test.
Pack lives near full	About 80% of time above 0.9	Cap resting charge. Test in the Lab first.
Resistance uniform	3.66–4.48 mΩ, ranges overlap	No action. Rules out a false alarm.
Balance acceptable	Resting spread 22 mV, worst 30 mV	Recheck if cells 1 or 2 drift.
Life about 1,620 cycles	Range 1,503–1,700; about 5 years at 0.88/day	Use the range; re-run with more logs.

Notice what is not there: no instruction to replace anything today. The tool buys you time and sharpens your question.

CHAPTER 17

Evidence, limits and honesty

What was tested, where the tool is strong, and what it does not claim.

We would rather lose an argument today than your trust next year.

The public-data test: Sandia and HUST

95 LFP cells from two university lab datasets, all A123 18650. Task: from one measurement at 100 full cycles, forecast the whole health curve. Scored on cells never used to set up the engine.

Result	Number
Engine vs plain physics formula, typical error	2.4–3.4 vs 18–21 points
On unfamiliar cells (Sandia): health-only-falls vs flexible version	3.0 vs 20.4 points off
Same test on HUST	3.5 vs 5.7 points off
Before the knee (health above 85%)	1.4 vs 3.4 points off (health-only-falls wins)
After fade speeds up (below 85%)	2.5 vs 3.3 points off (flexible wins)
Which Sandia cells reach end of life (flexible vs health-only-falls)	15 of 18 vs 5 of 18

The engine runs both versions and picks by situation. The Virtual Lab has also reproduced the published findings of leading peer-reviewed battery researchers, using their own experimental data.

Before you quote any number

- **Lab data is not field data.** No field pack has yet been checked against a measured capacity test or a real end of life. Blind predictions, filed before cells fail, are next.
- **LFP only.** Trained on one chemistry; the chemistry input does nothing yet.
- **Ranges are too tight.** They caught the true answer 12–25% of the time against a 90% target. We are fixing this. Treat them as minimum uncertainty.
- **Resistance at 25 °C is an estimate** when the log runs warm. Read the range.
- **Labels matter.** Defaults for age and rated cycles weaken every life number.

The validation report is public. Run the app on your own packs and try to break it. We want to hear when it fails.

CHAPTER 18

Your first hour with the app

Everything in this book, pointed at your own pack.

Open the live app

batteryanalytics.bayesiancybersecurity.com

Minutes	Do this	Business question answered
0–5	Log in, choose Quick, upload one 12–24 h log. Let the converter map unfamiliar columns.	Is it fine? How long? How much energy? (Ch 2)
5–10	Enter age, rated cycles, capacity, chemistry. Run again.	Can I quote these numbers? (Ch 12)
10–15	Do the warranty sum from the tiles.	Is my warranty safe? (Ch 5)
15–30	Tap **See detailed analytics →**. On Pack overview, flip through the eight map layers. Click the cell that stands out twice.	Which cell, and do I send someone? (Ch 6)
30–40	Pack overview: ten checks. Data & conversion: coverage.	Is the data honest? (Ch 11)
40–50	Life forecast tab, then Virtual Lab tab: vary temperature and resting charge.	Which habit pays? (Ch 9)
50–60	Write three actions with owners and dates.	What do I do now? (Ch 16)

Habits that make you good at this

- Start with the question, not the screen.
- Read the range before the number.
- Look for two signals before raising an alarm.
- When a check fails, ask "sensor, setting or tampering?" before "battery?".
- Run again with every new log. Trends beat any single reading.
- If the app says "not resolved", believe it. That sentence saves money.

> **One last thing.** If you have a battery question this book does not answer, tell us. That is how the next chapter gets written.

APPENDIX

Glossary and learn more

The jargon, in plain words.

Term	In plain words
BMS / CAN	The battery management system, and the network that carries its messages.
SoH / SoC / DoD	State of health (capacity vs new), state of charge (how full), depth of discharge (how deep a swing).
End of life (EOL) / RUL	When health falls to 80% (default). Remaining useful life is the cycles or years to get there.
Internal resistance (DCIR)	How hard a cell resists current. From the voltage jump when current changes, corrected to 25 °C.
Likely range	The span the true value probably falls in.
Charging peak	The voltage where a cell takes in charge fastest. Position, width and height are early health signs.
Lithium loss (LLI)	Lithium that no longer carries charge. Main fade type in LFP.
Active material loss (LAM)	Electrode material that stops working.
SEI	A film on the anode that grows with time and heat.
Settling time	How long voltage takes to recover after current stops.
Ten checks (MP1–MP10)	Physics sanity checks on the log.
Stuxnet-style attack	An attack that damages equipment while feeding operators believable "normal" data.
CAN spoofing	Sending false but well-formed messages on a CAN bus, which has no sender check.
MW distance	How unusual a cell or pack looks vs healthy LFP cells. Above about 3 is unfamiliar.
Knee	Where fade speeds up.
Quick / Detailed	The one-page summary / the full engineering view.
Resolved / NOT resolved	Whether an effect is bigger than the uncertainty.

Learn more

The live app	batteryanalytics.bayesiancybersecurity.com
Validation report (Sandia, HUST)	batteryanalytics.bayesiancybersecurity.com/static/MW_Monotone_Verdict_Sandia_HUST.pdf
Contact	Bayesian Cybersecurity Pvt Ltd, Thane